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Diffusion-PETS: Diffusion Model for Periodicity-Enhanced Time-Series Generation --Manuscript Draft--

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Abstract:	Reliable synthetic sensor time-series generation is important for Cyber-Physical Systems (CPS), where limited observations, sensor drift, and intrinsic uncertainty constrain data-driven modeling. Although diffusion models provide a principled framework for uncertainty-aware synthesis, many existing approaches do not explicitly represent interacting periodic structure within the denoising architecture. This limitation is particularly relevant in multi-period and longer-sequence settings. We propose Diffusion-PETS, a Transformer-based diffusion framework with an explicit periodic inductive bias. A ranked spectral bank is transformed into period-aligned two-dimensional branches, while a differentiable prefix-mask mechanism learns the number of active branches and retains hard selection in the forward pass. Period-aware TimesBlocks are inserted after the encoder and immediately before the final decoder block. For unconditional generation, PETS-U-AA applies frozen Anti-Anchor aggregation during sampling; for probabilistic forecasting, PETS-F uses history-conditioned period routing, confidence gating, and reconstruction guidance without accessing future targets during sampling. Experiments on ETTh, AirQuality, and Energy demonstrate competitive unconditional generation with dataset- and sequence-length-dependent gains. PETS-F improves both point-forecast and probabilistic scores on ETTh and held-out AirQuality, whereas Diffusion-TS remains stronger on Energy, indicating that the benefit depends on the stability of the available periodic evidence. Structural diagnostics further show improved preservation of periodic and local temporal patterns.

Dear Editor,

We would like to submit our manuscript entitled “**Diffusion-PETS: Diffusion Model for Periodicity-Enhanced Time-Series Generation**” for consideration for publication in *Neurocomputing*.

This paper proposes *Diffusion-PETS*, a diffusion-based time-series modeling framework that explicitly incorporates periodic inductive bias into the generative process. While recent diffusion models have demonstrated strong performance in sequential data modeling, they typically rely on generic architectures that require periodic structure to be learned implicitly from data. This limitation becomes particularly critical in real-world sensor data, where multiple interacting periodic patterns dominate system dynamics, especially in long-horizon settings.

To address this, our approach integrates (i) multi-period two-dimensional representations and (ii) a learnable top- k frequency selection mechanism to explicitly model interacting temporal cycles. These components are incorporated within a Transformer backbone using a selective $E + \text{before-last } D$ configuration, enabling efficient global temporal coherence while preserving stable denoising behavior.

Extensive experiments on diverse benchmark datasets demonstrate that Diffusion-PETS maintains robust performance under multi-period dynamics and provides clear advantages in long-horizon generation and forecasting. Furthermore, interpretability analysis reveals how the model progressively refines temporal structure and adaptively captures dominant periodic components during the diffusion process.

We believe this work aligns well with the scope of *Neurocomputing*, as it advances neural network architectures and generative modeling for structured sequential data, with practical relevance to cyber-physical systems and real-world time-series applications.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to *Neurocomputing*.

Thank you for your consideration.

Sincerely,
Jin Park (on behalf of all authors)

Abstract

Reliable synthetic sensor time-series generation is important for Cyber-Physical Systems (CPS), where limited observations, sensor drift, and intrinsic uncertainty constrain data-driven modeling. Although diffusion models provide a principled framework for uncertainty-aware synthesis, many existing approaches do not explicitly represent interacting periodic structure within the denoising architecture. This limitation is particularly relevant in multi-period and longer-sequence settings.

We propose Diffusion-PETS, a Transformer-based diffusion framework with an explicit periodic inductive bias. A ranked spectral bank is transformed into period-aligned two-dimensional branches, while a differentiable prefix-mask mechanism learns the number of active branches and retains hard selection in the forward pass. Period-aware TimesBlocks are inserted after the encoder and immediately before the final decoder block. For unconditional generation, PETS-U-AA applies frozen Anti-Anchor aggregation during sampling; for probabilistic forecasting, PETS-F uses history-conditioned period routing, confidence gating, and reconstruction guidance without accessing future targets during sampling.

Experiments on ETTh, AirQuality, and Energy demonstrate competitive unconditional generation with dataset- and sequence-length-dependent gains. PETS-F improves both point-forecast and probabilistic scores on ETTh and held-out AirQuality, whereas Diffusion-TS remains stronger on Energy, indicating that the benefit depends on the stability of the available periodic evidence. Structural diagnostics further show improved preservation of periodic and local temporal patterns.

Highlights

- Period-aligned two-dimensional branches expose interacting temporal cycles
- A learnable mask adapts the number of dominant periods used by the model
- Selective placement adds period awareness with limited spectral overhead
- Frozen branch aggregation and history-only routing tailor sampling by task
- Experiments show strong long-horizon synthesis and dataset-dependent forecast gains

Response to Reviewers

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General Response

Dear Editor and Reviewers,

We sincerely thank the Editor and both Reviewers for their careful reading of our manuscript and for the constructive comments. The revision has substantially strengthened both the methodological presentation and the experimental validation of Diffusion-PETS.

The principal revisions are as follows. First, we added time-series-specific diffusion preliminaries that distinguish chronological time from diffusion time and separate true-unconditional generation from conditional probabilistic forecasting. Second, we reorganized the method around a shared period-aware backbone and two task-specific protocols, PETS-U and PETS-F. Third, we clarified the unconditional-generation terminology: PETS-U denotes the model using batch-shared full-sequence routing and the canonical spectral-weighted aggregation, whereas PETS-U-AA denotes the same frozen PETS-U checkpoint sampled with Anti-Anchor applied symmetrically at the encoder- and decoder-side TimesBlocks. The location-specific labels Encoder-AA and Decoder-AA are used only in the corresponding ablation. Fourth, PETS-F now uses causally aligned history-conditioned routing, a fixed sample-wise history period bank, confidence-gated sampling, and reconstruction-guided reverse diffusion. Fifth, we added controlled analyses of representation strategy, TimesBlock placement, forecasting period routing, confidence gating, and feature fusion. The representation and placement studies use the same canonical aggregation across variants so that each requested experimental factor is isolated. Finally, we expanded the probabilistic-forecasting metrics and figure captions and narrowed the evaluated scope to true-unconditional generation and probabilistic forecasting.

We also moderated claims where the results reveal dataset-, sequence-length-, or metric-dependent trade-offs. Below, each Reviewer comment is reproduced, followed by our point-by-point response and the corresponding changes in the revised manuscript.

Reviewer #1

Comment 1

Reviewer comment

In various parts of the manuscript the authors focus on making comparison to a model Diffusion-TS proposed in the conference paper DIFFUSION-TS: INTERPRETABLE DIFFUSION FOR GENERAL TIME SERIES GENERATION by Xinyu Yuan and Yan Qiao. This makes an impression that Difusion-PETS is a slightly enhanced forked Diffusion-TS model. Please clearly state the differences between the two methods.

Response

We thank the Reviewer for pointing out that the original presentation could give the impression that Diffusion-PETS is merely an incremental modification of Diffusion-TS. We revised the Related Work and Method sections to make the methodological distinction explicit.

Diffusion-TS provides a strong Transformer-based diffusion baseline with time-series decomposition and frequency-domain objectives. Diffusion-PETS instead introduces an explicit period-aware architectural inductive bias within the denoising network. Its main components are: (1) data-dependent period-aligned 2D representations that expose within-cycle and across-cycle relations; (2) prefix-mask STE learnable-k selection that adapts the number of active period branches; (3) selective TimesBlock placement after the encoder and immediately before the final decoder block; and (4) task-specific routing and generation-time control of multi-period information.

For true-unconditional generation, PETS-U uses batch-shared full-sequence routing and is trained with the canonical spectral-weighted aggregation. The trained checkpoint is then frozen, and PETS-U-AA replaces only the generation-time branch aggregation at the two TimesBlocks with Anti-Anchor. Anti-Anchor retains non-opposing complementary period information while suppressing the component that opposes the dominant harmonic anchor; it changes neither period discovery nor the trained parameters. PETS-F addresses a different information setting through history-conditioned training, a fixed sample-wise history period bank, confidence-gated sampling, and reconstruction guidance.

Diffusion-TS is therefore retained as a directly comparable baseline, while the contribution of Diffusion-PETS lies in the representation, adaptive spectral capacity, selective placement, routing, residual fusion, and task-specific aggregation of multi-period information.

Changes in the revised manuscript

We revised Sec. 3.1, Table 1, and Secs. 4.1-4.4. Sec. 4.3.3 and Algorithm 1 now define PETS-U and PETS-U-AA explicitly, and Table 7 reports the encoder-decoder Anti-Anchor ablation.

Comment 2

Reviewer comment

What may strike the reader is not only the affinity of the solutions, but also the structure, presentation methods, visualizations, scores, etc. It looks like the manuscript is completely driven by the paper by Xinyu Yuan and Yan Qiao.

Response

We appreciate this concern. The original organization and the use of common benchmark conventions could make the presentation appear overly aligned with Diffusion-TS. We therefore reorganized the manuscript around the proposed period-aware method rather than around the baseline.

The revised manuscript introduces time-series-specific preliminaries, separates the shared period-aware backbone from the PETS-U and PETS-F protocols, and provides independent algorithms for unconditional generation and conditional forecasting. We also added analyses specific to Diffusion-PETS: an encoder-decoder aggregation study, a controlled frequency-representation comparison, history-only and sample-wise period routing, confidence-gated sampling, TimesBlock-placement analysis, structural-fidelity diagnostics, and feature-fusion analysis.

We retain established generative metrics because they enable evaluator parity and direct comparison, not because the proposed method is derived from the baseline. The revised manuscript defines these metrics

explicitly and supplements them with period-structure, calibration, probabilistic-forecasting, and mechanism-specific analyses.

Changes in the revised manuscript

We substantially revised Secs. 2-5, Algorithms 1-2, Tables 7-12, and the captions of the main figures. The manuscript is now organized around the shared period-aware backbone and the PETS-U/PETS-F task protocols.

Comment 3

Reviewer comment

The manuscript should provide some background information on application diffusion models to time series generation. The description in section 3.1 is too generic and can be applied to image generation as well. Generation of time series is a specific problem with two time domains. The manuscript should also present the distinction between prediction and generation of time series with specific features and patterns and discuss the scores that can be used to assess quality. E.g. these 90% confidence intervals appearing in Figs. 16-19 within DIFFUSION-TS: INTERPRETABLE DIFFUSION FOR GENERAL TIME SERIES GENERATION.

Response

We thank the Reviewer for this important suggestion. The preliminary material was rewritten specifically for time-series diffusion.

Sec. 2.1 now distinguishes chronological time, which indexes the physical evolution of a multivariate sequence, from diffusion time, which indexes the noise level applied to the complete sequence. A reverse transition from diffusion step t to $t-1$ therefore refines all chronological positions and variables simultaneously and does not move backward along the physical time axis.

Sec. 2.2 distinguishes true-unconditional generation from conditional forecasting. Unconditional generation maps Gaussian noise to a complete synthetic trajectory, whereas probabilistic forecasting generates a distribution over the future conditioned on an observed history. Table 3 defines the unconditional-generation metrics, and Table 5 defines ensemble-mean MAE/RMSE, expected-sample MAE, empirical CRPS, normalized Energy Score, and Weighted Interval Score. Figure 5 visualizes 100 stochastic trajectories, their pointwise ensemble mean, and central 50% and 90% prediction intervals.

Changes in the revised manuscript

We added/revised Secs. 2.1-2.2, Tables 3 and 5, and Fig. 5 to provide time-series-specific diffusion background, task definitions, and probabilistic evaluation.

Comment 4

Reviewer comment

In Section 4.2 a method of k -top frequency extraction is described. As I understand a batch of observations, consisting of a window and prediction horizon, is taken. The stride is equal to 1. In the next step, a spectrogram is computed, summed along the time axis, and the top k frequencies are selected. The problem is that with batch size equal to 128, and sizes of window and prediction horizon like 32, 64, 72... the predictor sees the future leaked by FFT frequencies. This may produce the excellent prediction results presented in Fig. 4. In the ablation study, please include a variant in which the top k frequencies are taken from the window only.

Response

We agree that using the future target to select forecasting-time period candidates would constitute information leakage. PETS-F was therefore revised so that its period-routing policy is aligned with the information available at the forecast origin. We also clarify that candidate selection uses a one-dimensional non-DC rFFT salience spectrum, not a spectrogram.

During PETS-F training, each TimesBlock constructs a sample-wise ranked candidate bank of capacity $k_{\max}$ from the first H positions of its own intermediate latent input. The checkpoint-learned STE count of that TimesBlock determines the active prefix. The selected period branches still transform the complete 128-step latent representation, and the diffusion and FFT objectives remain defined over the complete training

window, which is entirely contained in the chronological training partition. The active branch count is learned through a corrected prefix-mask

straight-through estimator: the forward pass uses a binary prefix over the ranked $k_{\max}$ -candidate bank, whereas the backward pass follows a temperature-controlled soft prefix boundary. The masked spectral weights are subsequently renormalized over the active slots.

At inference, SH64/CG64 computes the ranked period bank once from the observed 64-step history only. The encoder and decoder use the active prefixes determined by their checkpoint-learned STE counts, and this history-derived bank remains fixed throughout reverse diffusion. Neither the future target nor later observations are used for period selection or confidence estimation.

Table 10 provides the requested routing ablation. Under batch-shared routing, replacing full-sequence FFT (BF) with history-only FFT (BH) improves ensemble-mean MAE from 0.0867 to 0.0834 and empirical CRPS from 0.0734 to 0.0706. Sample-wise history-only routing (SH64) further improves these values to 0.0825 and 0.0698, respectively. The forecasting gain therefore does not rely on future-dependent period selection.

Changes in the revised manuscript

We revised Sec. 4.2.1 to describe the corrected prefix-mask STE learnable-k mechanism and revised Secs. 4.4.1--4.4.2 and Algorithm 2 to make the causal routing, $k_{\max}$ -candidate bank, and checkpoint-learned active prefix explicit. Table 10 reports the BF/BH/SH64 routing ablation.

Comment 5

Reviewer comment

In section 5.2: what do you mean by stating that "We note that MAE is computed differently for deterministic and probabilistic models." I wonder how much sense makes computation of MAE/MSE for generated time series. Typically these metrics smooth the predictions, whereas generation aims at recreating patterns comprising peaks, noise, etc. It would probably make sense to generate 100 trajectories, average them over time, and then compute MAE.

Response

We agree that MAE or MSE computed from one arbitrarily selected stochastic trajectory does not adequately characterize a probabilistic forecaster. The evaluation was revised to distinguish the predictive center, individual stochastic trajectories, and the complete predictive distribution.

Ensemble-mean MAE and RMSE are computed after averaging the generated trajectories pointwise at each forecast coordinate. The averaging is therefore performed across stochastic trajectories, not across chronological time. Expected-sample MAE evaluates the expected error of an individual trajectory. Empirical CRPS, normalized Energy Score, and Weighted Interval Score evaluate complementary properties of the predictive distribution, including marginal and joint fidelity, spread, calibration, coverage, and sharpness.

The aggregate FINAL_TEST comparison states the number of trajectories used per model seed. Figure 5 separately overlays 100 trajectories for qualitative inspection, together with their pointwise ensemble mean and central 50% and 90% prediction intervals, allowing peaks, phase, local variability, and uncertainty to be inspected directly.

Changes in the revised manuscript

We revised Sec. 5.3 and Table 5 to distinguish ensemble-mean, sample-wise, and distributional metrics. Table 6 states the FINAL_TEST repetition protocol, and Fig. 5 provides the 100-trajectory qualitative visualization.

Comment 6

Reviewer comment

Provide a more extensive description of the content presented in the figures. Preferably in their captions. For example, Fig. 3 Data Distribution visualization - what is actually visualized in the upper row? Vectors in prediction horizon after dimensionality reduction?

Response

We thank the Reviewer for identifying this ambiguity. The figure captions were expanded to state the input data, representation, colors, rows, and purpose of each panel.

The revised data-distribution figure (Fig. 4) states that the upper row contains two-dimensional t-SNE projections of TS2Vec embeddings computed from complete real and unconditionally generated T-step sequences, with red denoting real data and blue denoting generated data. The lower row contains kernel-density estimates of normalized scalar values pooled across temporal positions and variables from the same sample sets. Figure 3 now explicitly identifies PETS-U-AA and canonical PETS-U in the structural-fidelity comparison. The interpretability figures likewise identify the selected periods, denoising snapshots, and spectra shown at each reverse step.

Changes in the revised manuscript

We expanded the captions of Figs. 3-7. Fig. 3 names PETS-U-AA and PETS-U explicitly, while Fig. 4 defines the TS2Vec/t-SNE and KDE panels.

Reviewer #2

Comment 1

Reviewer comment

The motivation for adopting multi-period 2D representation is not sufficiently elaborated. The authors compare this scheme with spectrogram and frequency-as-channel methods, but do not quantitatively analyze their respective pros and cons on multi-period CPS time series.

Response

We thank the Reviewer for pointing out that the original manuscript discussed alternative frequency representations mainly at a conceptual level. We added a controlled ETTh-128 representation ablation.

The proposed Period-Aligned 2D representation is compared with complex STFT/Spectrogram and complex Frequency-as-Channel representations under the same true-unconditional PETS-U backbone. The Transformer architecture, E + before-last D placement, prefix-mask STE learnable-k, routing scope, diffusion and spectral objectives, checkpoint rule, evaluator protocol, and canonical spectral-weighted aggregation are held fixed so that representation is the only experimental factor. The trainable parameter counts differ by less than 0.2%, and each configuration is evaluated using three independent model seeds and five stochastic generation draws per seed.

Table 8 shows that Period-Aligned 2D obtains the lowest Correlational, Discriminative, Predictive, and Period-JSD scores. Frequency-as-Channel obtains the lowest Context-FID and spectral-fidelity error, while STFT/Spectrogram performs better in curvature and turning-point fidelity. The revised manuscript therefore presents a measured quality trade-off rather than universal superiority.

Changes in the revised manuscript

We expanded Secs. 3.2, 4.2.1, and 5.4, and added the controlled comparison in Table 8. Its caption states that the canonical spectral-weighted aggregation is shared across variants.

Comment 2

Reviewer comment

It is unclear why 2D period-aligned reshaping is more suitable for interacting temporal cycles than other frequency transformation strategies. Please add quantitative analysis or empirical comparison to strengthen this motivation.

Response

We agree that this motivation required empirical support. In Table 8, Period-Aligned 2D obtains a Period-JSD of 0.015 ± 0.002 , compared with 0.024 ± 0.007 for STFT/Spectrogram and 0.043 ± 0.004 for Frequency-as-Channel, corresponding to reductions of approximately 38% and 65%, respectively. It also obtains the best Correlational, Discriminative, and Predictive Scores among the three representations.

We interpret this result through representation geometry. STFT organizes features by local time and frequency, while Frequency-as-Channel augments the one-dimensional temporal representation with global spectral coefficients. Period-Aligned 2D reorganizes the latent sequence according to each selected period, explicitly exposing repeated-cycle and within-cycle phase coordinates to the 2D feature extractor. This makes interactions within and across repeated cycles directly accessible to the convolutional operator.

The claim was also moderated. Frequency-as-Channel is better on Context-FID and spectral fidelity, and STFT is better on several local-shape metrics. We therefore describe Period-Aligned 2D as a favorable inductive bias for period-distribution preservation and interacting periodic dependencies in the controlled ETTh-128 setting, not as a universally superior frequency transform.

Changes in the revised manuscript

We revised the conceptual motivation and limitations in Secs. 3.2 and 4.2.1. The controlled empirical evidence is reported in Table 8 and discussed in Sec. 5.4.

Comment 3

Reviewer comment

The proposed model is applied to generation, forecasting and imputation tasks, but the challenges of periodic modeling under conditional constraints are rarely analyzed.

Response

We thank the Reviewer for highlighting that periodic modeling under conditional constraints requires a different treatment from true-unconditional generation. We introduced a separate PETS-F formulation and expanded the forecasting analysis.

The revised method identifies three constraints: period evidence must be derived only from information available at the forecast origin; a finite historical spectrum can be weak or ambiguous; and generated samples must remain consistent with observed values. PETS-F addresses these constraints through history-conditioned training, a fixed sample-wise history period bank (SH64), a parameter-free confidence gate (CG64), and reconstruction-guided reverse diffusion.

Table 10 quantifies history-only and sample-wise routing, and Table 11 isolates the confidence gate after the PETS-F backbone and SH64 policy are fixed. The forecasting result is explicitly reported as dataset-dependent: PETS-F improves the final comparison on ETTh and the held-out confirmatory AirQuality FINAL_TEST, whereas Diffusion-TS remains stronger on Energy.

To avoid overextending the scope, the previous imputation claims were removed and the revised manuscript focuses on true-unconditional generation and probabilistic forecasting.

Changes in the revised manuscript

We expanded Sec. 4.4 and Sec. 5.3, added the routing and confidence-gating analyses in Tables 10-11, and narrowed the evaluated scope to unconditional generation and probabilistic forecasting.

Comment 4

Reviewer comment

The authors claim that inserting period-aware TimesBlock only at the encoder output and penultimate decoder layer balances performance and computation, but do not explain why inserting TimesBlock into other positions (e.g., all encoder/decoder layers, middle decoder layers) causes performance degradation.

Response

We thank the Reviewer for requesting direct empirical support for TimesBlock placement. We added a focused ETTh-128 true-unconditional PETS-U placement ablation.

Three trained configurations are compared: E-only, E + MidD, and the proposed E + LastBeforeD. E + MidD and E + LastBeforeD contain the same two TimesBlocks and have identical parameter counts, isolating the decoder insertion location. All trained variants share the Transformer backbone, TimesBlock implementation, prefix-mask STE learnable-k policy, routing, objective, checkpoint rule, evaluators, and canonical spectral-weighted aggregation, so placement is the only generation-quality factor.

The results reveal a metric-dependent trade-off. E + LastBeforeD obtains lower Correlational and Predictive Scores than E + MidD, whereas E + MidD obtains lower Context-FID and Discriminative Scores. Relative to E-only, adding the before-last decoder TimesBlock improves Correlational, Discriminative, and Predictive Scores while increasing Context-FID. We therefore describe the proposed location as a selective late-refinement design rather than as universally optimal.

An All-Layers configuration with six independent TimesBlocks was additionally instantiated only for forward-cost profiling. Relative to the proposed two-block model, it increases parameters from 1.592M to 3.888M and denoiser-forward latency from 30.376 ms to 72.517 ms. It was not trained or ranked on generation quality.

Changes in the revised manuscript

We revised Sec. 4.2.2 and Sec. 5.4, and added the placement and forward-cost analysis in Table 9. The caption states that the canonical spectral-weighted aggregation is shared across trained variants and that All-Layers is cost-profile-only.

Comment 5

Reviewer comment

Self-attention captures local and long-range temporal dependencies, while TimesBlock extracts periodic structures. The authors do not analyze the feature fusion mechanism between these two modules.

Response

We thank the Reviewer for this comment. Sec. 4.2.2 now makes the fusion operation explicit: TimesBlock does not replace the attention-derived representation but produces a period-aware residual that is added element-wise to the Transformer feature at both period-aware locations.

Sec. 5.5 and Table 12 quantify this interaction using the residual-to-base magnitude ratio r and cosine similarity c . Across three model seeds, the encoder-side fusion yields $r = 0.6349 \pm 0.0373$ and $c = -0.2553 \pm 0.0209$, whereas the decoder-side fusion yields $r = 0.4198 \pm 0.0451$ and $c = -0.3412 \pm 0.0408$. The encoder therefore introduces a larger relative modification, while the smaller but more negatively aligned decoder residual acts as a more directionally corrective late refinement. The consistently negative cosine similarities also indicate that the TimesBlock residual is not redundant with the contextual feature produced by self-attention.

Changes in the revised manuscript

We corrected the residual-fusion equations in Sec. 4.2.2 and added the feature-contribution analysis in Sec. 5.5 and Table 12.

Comment 6

Reviewer comment

The introduction needs to be further improved, and what are the different of the proposed designed in this paper, compared with existing model-based scheme, such as <https://ieeexplore.ieee.org/document/11397770> <https://spj.science.org/doi/10.34133/research.0631>

Response

We thank the Reviewer for this suggestion. The Introduction and Related Work were expanded to distinguish data-driven generative diffusion from model-based CPS schemes.

Model-based approaches encode governing equations, identified dynamics, structural constraints, or known physical parameters and can provide strong interpretability and extrapolation when an accurate system model is available. Diffusion-PETS instead learns a stochastic distribution of multivariate sensor trajectories directly from data and introduces periodicity as an architectural prior without assuming an explicit plant model. It is intended for settings in which periodic behavior is observable but a complete mechanistic model is unavailable or incomplete.

We also clarified the distinction from decomposition-based and operator-based temporal methods such as Autoformer, FEDformer, FNO, and TimesNet. These models are primarily designed for deterministic forecasting or operator learning, whereas Diffusion-PETS models stochastic trajectories and predictive

distributions through iterative denoising. The model-based works suggested by the Reviewer are therefore used to sharpen the scope distinction rather than treated as direct generative baselines. The proposed periodic bias is positioned as a data-driven structural prior that complements, rather than replaces, physics- or model-based CPS approaches.

Changes in the revised manuscript

We examined both references suggested by the Reviewer. Their problem formulations concern model-based physical-system design rather than stochastic time-series generation; in particular, one reference addresses programmable-metasurface-based millimeter-wave OAM generation. We therefore do not treat these works as direct generative baselines. Instead, we expanded the Introduction and Discussion to clarify the distinction between model-based CPS simulation and data-driven stochastic time-series generation.

Diffusion-PETS: Diffusion Model for Periodicity-Enhanced Time-Series Generation^{*,**}

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ABSTRACT

Reliable synthetic sensor time-series generation is important for Cyber-Physical Systems (CPS), where limited observations, sensor drift, and intrinsic uncertainty constrain data-driven modeling. Although diffusion models provide a principled framework for uncertainty-aware synthesis, many existing approaches do not explicitly represent interacting periodic structure within the denoising architecture. This limitation is particularly relevant in multi-period and longer-sequence settings. We propose Diffusion-PETS, a Transformer-based diffusion framework with an explicit periodic inductive bias. A ranked spectral bank is transformed into period-aligned two-dimensional branches, while a differentiable prefix-mask mechanism learns the number of active branches and retains hard selection in the forward pass. Period-aware TimesBlocks are inserted after the encoder and immediately before the final decoder block. For unconditional generation, PETS-U-AA applies frozen Anti-Anchor aggregation during sampling; for probabilistic forecasting, PETS-F uses history-conditioned period routing, confidence gating, and reconstruction guidance without accessing future targets during sampling. Experiments on ETTh, AirQuality, and Energy demonstrate competitive unconditional generation with dataset- and sequence-length-dependent gains. PETS-F improves both point-forecast and probabilistic scores on ETTh and held-out AirQuality, whereas Diffusion-TS remains stronger on Energy, indicating that the benefit depends on the stability of the available periodic evidence. Structural diagnostics further show improved preservation of periodic and local temporal patterns.

1. Introduction

CPS increasingly depend on multivariate sensor time-series data to monitor, predict, and control complex physical processes. Such data arise from tight interactions between physical dynamics and cyber components in industrial automation, smart grids, environmental monitoring, and intelligent transportation systems. In practice, however, collecting high-quality sensor time series at scale is often constrained by sensor drift and calibration errors [1], limited deployment durations, privacy concerns, and the high cost of continuous monitoring. These constraints significantly hinder the development of data-driven CPS intelligence and motivate the need for reliable synthetic time-series generation as a complementary data source.

Recent progress in deep generative modeling has made realistic time-series synthesis increasingly feasible. In particular, score-based diffusion models [2], and notably denoising diffusion probabilistic models (DDPMs) [3], have

demonstrated strong generative performance with stable optimization dynamics. Unlike adversarial approaches, diffusion models avoid mode collapse and training instability while producing high-fidelity samples. Their success across images [4, 5], audio [6], video [7, 8, 9], and text [10, 11, 12] has naturally extended to time-series domains, including forecasting and imputation [13, 14, 15], where diffusion models effectively capture sequential structure and uncertainty.

The probabilistic formulation of diffusion models offers important advantages for CPS-oriented time-series analysis. CPS inherently operate under uncertainty due to sensor noise, partial observability, and environmental variability. DDPMs model data generation through a sequence of conditional probability distributions, allowing uncertainty to be propagated throughout the synthesis process. This property is well suited to CPS tasks such as forecasting, fault analysis, predictive maintenance, and scenario simulation, where decisions must account for plausible deviations from nominal behavior. Moreover, the reverse diffusion process performs uncertainty-aware iterative refinement, resembling state-estimation procedures in CPS simulation and digital-twin modeling.

Despite these advantages, real-world sensor time series pose additional challenges due to complex and interacting periodic structures. Such behavior is commonly observed across a wide range of domains, including air-quality monitoring [16], energy systems [17], traffic networks [18], gas sensor arrays [19], and other large-scale sensing infrastructures. Fig. 1 shows the distributions of the

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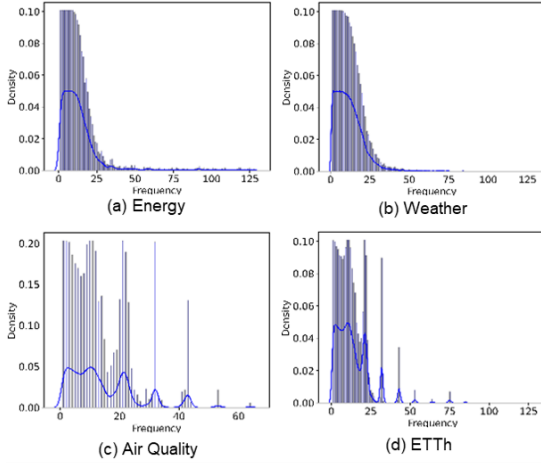


Figure 1: Distributions of the 50 largest non-DC frequency magnitudes for representative datasets: (a) Energy, (b) Weather, (c) AirQuality, and (d) ETTh. Energy and Weather are dominated by low frequencies, whereas AirQuality and ETTh exhibit broader spectral profiles.

50 largest non-DC frequency magnitudes across representative datasets. In the first row (Energy and Weather), the spectra are strongly low-frequency dominated, whereas the second row shows broader and more irregular frequency distributions. These data often exhibit multiple overlapping periodicities across daily, weekly, and seasonal scales, whose interactions are nonlinear and time-varying, driven by environmental conditions and operational regimes. Capturing such multi-scale interactions is essential for faithful synthesis and long-horizon generation, yet remains challenging under limited training data.

Although diffusion models have recently been extended to time-series generation, most existing architectures rely on recurrent or attention-based backbones that lack explicit inductive bias for periodic structure [20, 21]. Recurrent networks must implicitly memorize cycles through hidden states, which becomes unreliable for long or interacting periods, while standard Transformers treat time primarily as positional indices without inherent frequency awareness [22]. Consequently, periodic behavior must be learned entirely from data, leading to inefficiencies and degraded generalization, particularly in multi-period settings with interacting temporal patterns.

To address these limitations, we propose Diffusion-PETS, a Transformer-based diffusion model that embeds periodic inductive bias directly into the generative process. Inspired by the core assumptions of TimesNet [23], Diffusion-PETS leverages (i) multi-period two-dimensional representations to disentangle overlapping temporal patterns, and (ii) dominant top- k frequency components to capture essential periodic structure. Here, a 2D representation denotes a period-aligned reshaping of a 1D sequence into a matrix indexed by phase-within-period and period index, making intra-period and inter-period variations explicit. The resulting multi-period 2D maps are then processed with

multi-scale 2D kernels—for example, using a parameter-efficient Inception-style block—to capture periodic dependencies across multiple receptive fields (Fig. 2(b)). This differs from spectrogram-based representations, which produce local time-frequency maps via sliding-window FFTs, and from frequency-as-channel approaches, which preserve the original temporal axis and append spectral information as channels or embeddings.

By integrating these principles into diffusion modeling, Diffusion-PETS constrains generation around physically meaningful rhythms, improving stability, interpretability, and long-horizon coherence. From a CPS perspective, this integration is particularly important, as realistic synthesis must respect underlying physical dynamics rather than producing visually plausible but structurally inconsistent signals. Diffusion-PETS should also be distinguished from model-based CPS schemes. Physics- or equation-based models simulate trajectories from identified dynamics and known parameters, whereas Diffusion-PETS learns a stochastic data distribution without requiring an explicit plant model. The proposed periodic inductive bias is therefore a data-driven structural prior rather than a replacement for physical system identification.

Our contributions can be summarized as follows.

- We propose Diffusion-PETS, a diffusion-based framework that embeds periodic inductive bias through multi-period 2D representations and a learnable top- k frequency selection mechanism, enabling adaptive modeling of complex periodic structures for time-series generation.
- We introduce a selective architectural integration strategy, placing period-aware modules after the encoder and immediately before the final decoder block to combine global period-aware context with late residual refinement while avoiding repeated spectral processing throughout the network.
- We introduce task-specific generation protocols: PETS-U-AA combines batch-shared full-sequence routing with a frozen sampling-time branch-aggregation rule, whereas PETS-F uses causal history-conditioned routing and probabilistic forecasting controls.

2. Preliminaries

2.1. Diffusion Framework for Time-Series Generation

A diffusion model for time-series generation transforms Gaussian noise into structured temporal sequences through iterative denoising. Unlike images, time series have an explicit chronological ordering and may exhibit trend, seasonality, periodicity, long-range dependencies, and interactions across variables. The denoising network must therefore restore both temporal and cross-variable structures.

Time-series diffusion involves two distinct temporal axes. Let $\mathbf{X}_0 = (\mathbf{x}_1, \dots, \mathbf{x}_L) \in \mathbb{R}^{L \times d}$ denote a clean

multivariate time series, where $n \in \{1, \dots, L\}$ indexes chronological time. In contrast, $t \in \{1, \dots, S\}$ indexes the diffusion process. The former describes the temporal evolution of the observed process, whereas the latter controls the noise level applied to the entire sequence. Accordingly, the transition from $\mathbf{X}_t$ to $\mathbf{X}_{t-1}$ does not move the sequence backward in chronological time; instead, it refines all chronological positions and variables simultaneously.

Given a noise schedule $\beta_t \in (0, 1)$, define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{r=1}^t \alpha_r$. The noisy time series at an arbitrary diffusion step is expressed as

$$q(\mathbf{X}_t | \mathbf{X}_0) = \mathcal{N}(\mathbf{X}_t; \sqrt{\bar{\alpha}_t} \mathbf{X}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (1)$$

or equivalently,

$$\mathbf{X}_t = \sqrt{\bar{\alpha}_t} \mathbf{X}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

The reverse process starts from $\mathbf{X}_S \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and progressively reconstructs the complete sequence:

$$p_\theta(\mathbf{X}_{t-1} | \mathbf{X}_t, \mathbf{c}) = \mathcal{N}(\mathbf{X}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{X}_t, t, \mathbf{c}), \boldsymbol{\Sigma}_\theta(\mathbf{X}_t, t, \mathbf{c})), \quad (3)$$

where $\mathbf{c}$ denotes optional conditioning information. For unconditional generation, $\mathbf{c} = \emptyset$, whereas for conditional generation it may include an observed history or mask. The denoising network may predict the added noise [3] or directly estimate the clean sequence and its structured components, as in Diffusion-TS [24]. Diffusion-PETS focuses on the denoising architecture by introducing adaptive period-aware representations for restoring periodic temporal structures.

2.2. Problem Statement

Let the observed dataset be

$$\mathcal{D} = \{\mathbf{X}_{1:L}^i\}_{i=1}^N, \quad \mathbf{X}_{1:L}^i \in \mathbb{R}^{L \times d}, \quad (4)$$

where N , L , and d denote the number of samples, sequence length, and number of variables, respectively.

For unconditional generation, the model $\mathbb{G}_u$ maps Gaussian noise to a complete synthetic time series:

$$\hat{\mathbf{X}}_{1:L}^i = \mathbb{G}_u(\mathbf{Z}^i), \quad \mathbf{Z}^i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (5)$$

For conditional forecasting, given an observed history $\mathbf{X}_{1:c}^i$ with $c < L$, the model $\mathbb{G}_c$ generates the future sequence as

$$\hat{\mathbf{X}}_{c+1:L}^i = \mathbb{G}_c(\mathbf{X}_{1:c}^i, \mathbf{Z}^i), \quad \hat{\mathbf{X}}_{c+1:L}^i \in \mathbb{R}^{(L-c) \times d}. \quad (6)$$

3. Related Work

3.1. Deep generative modeling of time series

Deep generative modeling of time series has been predominantly explored through GAN-based frameworks. Early work such as C-RNN-GAN [25] introduced recurrent generators for sequential synthesis. TimeGAN [26] represents a major milestone by combining adversarial and supervised objectives to jointly preserve temporal dynamics and distributional realism. Subsequent studies improved long-range dependency modeling using attention and transformer architectures. Despite their empirical success, GAN-based approaches remain challenged by training instability and mode collapse, particularly for long-horizon generation. VAE-based methods have also been investigated due to their probabilistic formulation and interpretable latent representations [27], but they often produce overly smooth samples, suffer from posterior collapse, and struggle to model complex long-range temporal dependencies, limiting their effectiveness for high-fidelity time-series generation.

Recent advances have increasingly shifted toward denoising diffusion probabilistic models, which offer stable training and strong uncertainty modeling. DiffWave [6], originally proposed for audio synthesis, demonstrated the effectiveness of diffusion models for structured sequential generation and inspired their adoption in time-series domains. Building on this foundation, TimeGrad [13] introduced conditional diffusion for time-series forecasting by integrating recurrent encoders, while CSDI [14] addressed imputation via transformer-based conditioning on observed values and masks. TimeDiff [15] further incorporated time-series-specific inductive biases, including partial future conditioning and autoregressive initialization.

More recently, unconditional diffusion models have been explored as task-agnostic frameworks for time-series generation. Diffusion-TS [24] combines a Transformer-based denoising network with seasonal-trend decomposition and frequency-domain objectives, demonstrating strong performance across general time-series generation tasks. Although Diffusion-TS and Diffusion-PETS both exploit frequency-domain information, they differ fundamentally in how periodic structure is represented and incorporated into the denoising process, as summarized in Table 1. Diffusion-TS relies on predefined decomposition components and frequency-domain objectives, whereas Diffusion-PETS reorganizes latent features into data-dependent, period-aligned 2D branches and adaptively controls their spectral capacity through learnable hard top- k selection. Diffusion-PETS further introduces selective encoder and before-last-decoder placement, together with generation-specific period-routing and branch-aggregation policies. Although PETS-F employs task-specific conditional sampling for forecasting, the core contribution shared by PETS-U and PETS-F lies in the period-aware representation and information-flow design of the denoising architecture, rather than in conditioning or objective modification alone. The proposed architecture is therefore complementary to existing conditioning,

Table 1

Architectural comparison of Diffusion-TS and Diffusion-PETS.

Aspect	Diffusion-TS	Diffusion-PETS
Period modeling	Decomposition / frequency-domain Fixed heuristic	Period-aligned 2D branches
Spectral capacity		Learnable hard top- k
Placement	Repeated baseline processing	Encoder output + before-last decoder
Uncond. routing	Native	Batch-shared full-sequence PETS-U
Uncond. aggregation	Native	Canonical training → Anti-Anchor sampling
Forecast routing	Native conditional sampler	History-conditioned PETS-F + SH64/CG64
Forecast residual control	Native conditional sampler	Fixed SH64 period bank, CG64 confidence scaling, and reconstruction guidance

masking-, and objective-based diffusion methods [13, 14, 15].

3.2. 2D representations of time-series using FFT

Constructing 2D representations of time-series using FFT has emerged as a powerful strategy to enrich temporal modeling by revealing periodic structures and cross-period interactions. Methodological variations typically fall into three categories: (1) Spectrogram-based representations, where time-series are windowed and FFT is applied locally to generate time-frequency matrices—commonly used in STFT-based models like STFT-GAN [28]; (2) Periodogram-based rearrangement, such as in Periodic Series Decomposition (PSD), where a signal is folded into a 2D matrix with rows representing periods (or harmonics) to capture intra- and inter-period interactions, as demonstrated in TimesNet [23]; and (3) Frequency-as-channel encoding, where FFT outputs are treated as channels or incorporated as positional encodings in temporal models, as used in Fourier Neural Operator [29] and FEDformer [30]. Each approach leverages frequency information differently—either by capturing localized spectral variations, enhancing temporal locality via reshaping, or augmenting neural networks with global periodic priors—thus offering diverse yet complementary inductive biases for time-series representation.

While several recent diffusion-based generative models transform time series into two-dimensional representations and perform diffusion in the resulting space [31, 32], they do not explicitly organize these representations around multiple data-dependent periodicities or model intra- and inter-period dependencies. Diffusion-PETS builds on period-aligned rearrangement by adaptively selecting dominant periods through top- k spectral analysis and constructing a structured 2D representation for each period candidate. These representations make within-period and across-period interactions explicit and integrate them directly into the denoising network, providing a periodicity-aware architectural inductive bias that is distinct from existing image-like 2D diffusion formulations. To complement this conceptual distinction, Sec. 5.4 quantitatively compares these three representation strategies under a common PETS-U backbone.

4. The proposed Diffusion-PETS method

4.1. Overview of Diffusion-PETS

Diffusion-PETS is a diffusion-based time-series generative framework that introduces an explicit periodic inductive bias into a Transformer denoising backbone. Its central idea is to complement attention-based temporal modeling with period-aligned representations that expose recurrent structure at multiple temporal scales. As illustrated in Fig. 2, the proposed framework consists of two shared architectural components: a learnable- k multi-period TimesBlock that adaptively selects and transforms dominant periodic modes, and a Periodic Inductive Bias Transformer (PIB-Transformer) that inserts this module only at selected stages of the denoising network. The resulting architecture preserves the flexibility of self-attention for local and stochastic dependencies while explicitly modeling longer-range periodic organization.

The same period-aware backbone supports two generation protocols with different information availability. For unconditional generation, **PETS-U** models complete sequences and determines period evidence from the full sequence representation. For probabilistic forecasting, **PETS-F** retains the same backbone and full-window diffusion objective but aligns period routing with the observed history; conditional generation is then performed using causal history-based sampling, as described in Sec. 4.4. This separation distinguishes the shared architectural inductive bias from task-specific routing and generation-time control. The TimesBlock transformation is shared, whereas the source of period evidence and the sampling-time control applied to the periodic residual are adapted to each generation task.

4.2. Shared Period-Aware Backbone

The shared backbone of Diffusion-PETS is designed around two complementary principles. First, periodic structure should be represented in a form that makes repeated cycles and within-cycle variation directly accessible to the feature extractor. Second, this period-aware transformation should be introduced selectively rather than replacing the local modeling capability of the Transformer. We realize these principles through the learnable- k multi-period TimesBlock and the PIB-Transformer shown in Fig. 2.

4.2.1. Learnable- k Multi-Period TimesBlock

Let $\mathbf{H} \in \mathbb{R}^{B \times T \times d}$ denote the latent representation entering a TimesBlock, and let $\mathbf{S}$ denote the temporal region used to estimate period evidence. The choice of $\mathbf{S}$ depends on the generation protocol and is specified in Secs. 4.3 and 4.4; the period-aware transformation described below is otherwise shared.

We first compute the real-valued Fourier spectrum of $\mathbf{S}$ along the temporal dimension. For each positive non-DC frequency f , its salience is obtained by averaging the amplitude over the corresponding routing group and latent channels:

$$A(f) = \text{Avg}_{\mathcal{R},c} \left| \mathcal{F}_t(\mathbf{S}_{:, :, c})_f \right|, \quad f = 1, \dots, \left\lfloor \frac{T_s}{2} \right\rfloor, \quad (7)$$

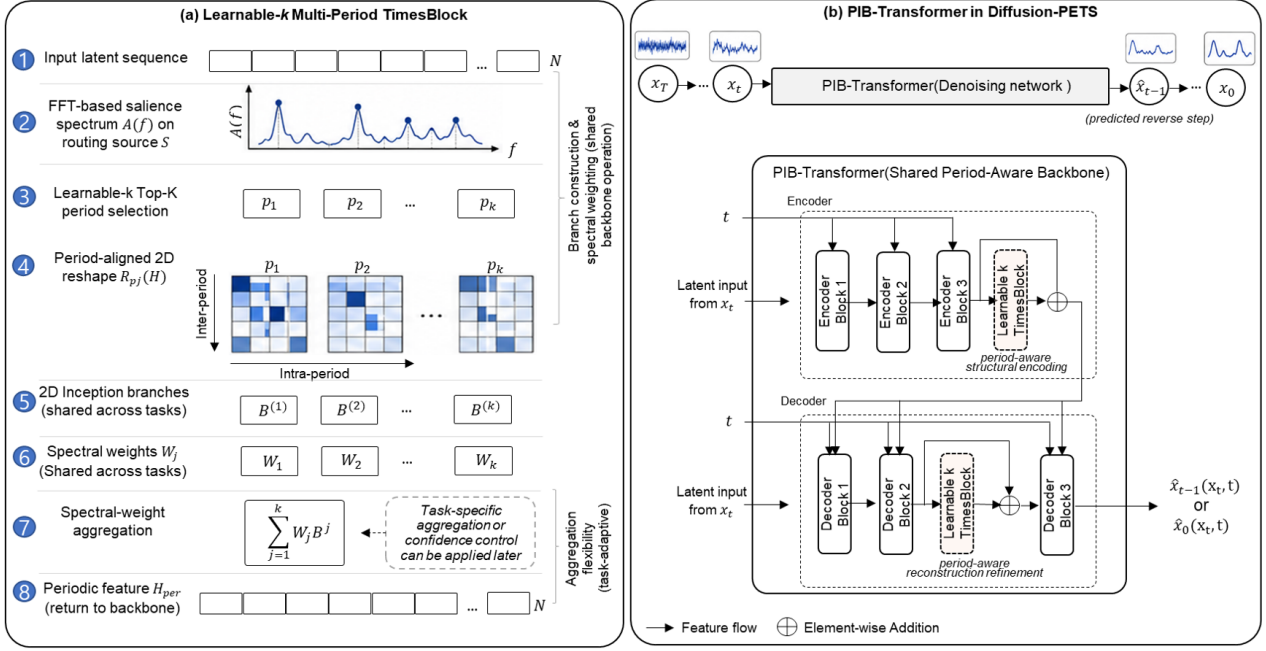


Figure 2: Shared period-aware backbone of Diffusion-PETS. (a) A learnable- k TimesBlock extracts dominant periods, constructs period-aligned 2D branches, and combines them using the canonical spectral-weighted aggregation; task-specific generation-time controls are described in Secs. 4.3 and 4.4. (b) The PIB-Transformer inserts learnable- k TimesBlocks after the encoder and immediately before the final decoder block, with their outputs fused through residual addition.

where T_s is the temporal length of $\mathbf{S}$ and $\mathcal{R}$ denotes the routing scope. Rather than changing the tensor shape through a discrete top- k operation, the TimesBlock first constructs a ranked candidate bank of fixed capacity $k_{\max}$:

$$\begin{aligned} \mathcal{I}_{\max} &= \text{TopK}(A, k_{\max}), \\ p_j &= \left\lceil \frac{T_s}{f_j} \right\rceil, \quad f_j \in \mathcal{I}_{\max}, \quad j = 1, \dots, k_{\max}. \end{aligned} \quad (8)$$

The candidates are ordered by decreasing spectral salience.

A fixed number of active branches would impose the same spectral capacity on time series with different periodic complexity. Each TimesBlock therefore learns a continuous capacity and its hard forward value as

$$\begin{aligned} k_{\text{cont}} &= k_{\min} + \sigma(\alpha)(k_{\max} - k_{\min}), \\ k_{\text{hard}} &= \text{round}(k_{\text{cont}}). \end{aligned} \quad (9)$$

Here, α is a trainable scalar specific to the TimesBlock. Applying an STE only to the rounded scalar does not make discrete branch membership differentiable. We therefore apply the STE to a prefix mask over the ranked candidate slots. For $j = 1, \dots, k_{\max}$,

$$\begin{aligned} m_j^{\text{hard}} &= \mathbb{I}(j \leq k_{\text{hard}}), \\ m_j^{\text{soft}} &= \sigma\left(\frac{k_{\text{cont}} - j + 0.5}{\tau_k}\right), \\ m_j^{\text{STE}} &= m_j^{\text{hard}} + m_j^{\text{soft}} - \text{sg}(m_j^{\text{soft}}). \end{aligned} \quad (10)$$

Here, τ_k is the mask temperature and $\text{sg}(\cdot)$ denotes stop-gradient. Thus, the forward pass uses an exactly binary prefix, whereas the backward pass follows the smooth boundary of the prefix mask.

For each candidate period p_j , the full TimesBlock input $\mathbf{H}$ is temporally padded when necessary and reshaped into a period-aligned two-dimensional representation,

$$\mathbf{U}^{(j)} = \mathcal{R}_{p_j}(\mathbf{H}) \in \mathbb{R}^{B \times \lceil T/p_j \rceil \times p_j \times d}, \quad (11)$$

where one axis indexes repeated cycles and the other indexes phase within a cycle. This organization exposes variations among phases within a cycle and among repeated cycles at the same phase. Each period-specific map is processed by a parameter-efficient multi-scale 2D convolutional block and restored to the original temporal layout:

$$\mathbf{B}^{(j)} = \mathcal{R}_{p_j}^{-1}[\text{Inception}_{2D}(\mathbf{U}^{(j)})]_{1:T}. \quad (12)$$

Let $s_j = A(f_j)$ be the salience score of the j -th ranked candidate. The prefix mask is applied to the branch weights before renormalization:

$$\begin{aligned} w_j &= \frac{m_j^{\text{STE}} \exp(s_j)}{\sum_{\ell=1}^{k_{\max}} m_{\ell}^{\text{STE}} \exp(s_{\ell}) + \epsilon}, \\ \mathbf{H}_{\text{per}} &= \sum_{j=1}^{k_{\max}} w_j \mathbf{B}^{(j)}. \end{aligned} \quad (13)$$

Consequently, inactive branches have exactly zero contribution in the forward pass, while the denoising objective can still update α through the smooth mask boundary.

More explicitly, let $g_j = \partial \mathcal{L} / \partial m_j^{\text{STE}}$ include the effect of the weight renormalization in Eq. (13). The capacity gradient is approximated by

$$\frac{\partial \mathcal{L}}{\partial \alpha} \approx \frac{k_{\max} - k_{\min}}{\tau_k} \sigma(\alpha) [1 - \sigma(\alpha)] \times \sum_{j=1}^{k_{\max}} g_j m_j^{\text{soft}} (1 - m_j^{\text{soft}}). \quad (14)$$

The factor $m_j^{\text{soft}} (1 - m_j^{\text{soft}})$ concentrates the gradient near the current prefix boundary. The model can therefore increase or decrease the effective number of active period branches while retaining hard prefix selection and spectral sparsity in the forward pass.

Equation (13) defines the canonical spectral-weighted aggregation used during training. PETS-F retains this aggregation during conditional sampling, whereas PETS-U-AA replaces only the frozen unconditional-sampling aggregation, as described in Sec. 4.3.3. The learnable prefix mask determines how many ranked candidates are active; the task-specific routing protocol determines the temporal evidence from which those candidates are ranked.

4.2.2. PIB-Transformer and Period-Aware Block Placement

Although self-attention provides flexible modeling of local and long-range dependencies, it does not explicitly distinguish temporal relations generated by repeated cycles. Conversely, inserting frequency-aware transformations throughout the network may over-constrain noisy intermediate representations and unnecessarily increase computation. The PIB-Transformer therefore adopts a selective integration strategy in which the TimesBlock is introduced at two stages only: after the Transformer encoder and immediately before the final decoder block. We refer to this architecture as the **E + before-last D** configuration.

Let the encoder be denoted by f_E , and let the decoder consist of L_D stacked blocks $\{D_\ell\}_{\ell=1}^{L_D}$. For compactness, we define $D_{1:\ell} = D_\ell \circ \dots \circ D_1$. For a noisy diffusion state $\mathbf{x}_t$, the encoder representation and its period-aware residual fusion are given by

$$\mathbf{h}^E = f_E(\mathbf{x}_t, t), \quad \tilde{\mathbf{h}}^E = \mathbf{h}^E + \text{TB}_E(\mathbf{h}^E). \quad (15)$$

Here, $\Delta \mathbf{h}_{\text{per}}^E \equiv \text{TB}_E(\mathbf{h}^E)$ is the encoder-side periodic residual. Thus, $\mathbf{h}^E$ retains the attention-derived contextual representation, while the TimesBlock provides an additive period-aware correction.

Let $\mathbf{h}_0^D$ denote the decoder input embedding. The first $L_D - 1$ decoder blocks process it using the encoder memory $\tilde{\mathbf{h}}^E$ and diffusion timestep t :

$$\mathbf{h}_{L_D-1}^D = D_{1:L_D-1}(\mathbf{h}_0^D; \tilde{\mathbf{h}}^E, t). \quad (16)$$

Immediately before the final decoder block, the decoder-side TimesBlock is fused through the same residual mechanism:

$$\tilde{\mathbf{h}}_{L_D-1}^D = \mathbf{h}_{L_D-1}^D + \text{TB}_D(\mathbf{h}_{L_D-1}^D). \quad (17)$$

Here, $\Delta \mathbf{h}_{\text{per}}^D \equiv \text{TB}_D(\mathbf{h}_{L_D-1}^D)$ denotes the late decoder-side periodic correction. The final decoder block produces the diffusion prediction as

$$\hat{\mathbf{e}}_\theta(\mathbf{x}_t, t) = D_{L_D}(\tilde{\mathbf{h}}_{L_D-1}^D; \tilde{\mathbf{h}}^E, t). \quad (18)$$

The encoder-side fusion introduces period-aware information into a globally contextualized latent representation, whereas the decoder-side fusion provides a late correction close to the diffusion prediction head. All intermediate Transformer blocks remain unchanged, preserving attention-based modeling of local and non-periodic dependencies. The relative strength and directional interaction of these two residual corrections are quantified in Table 12 using their residual-to-base magnitude ratios and cosine similarities.

The two insertion points are intended to provide complementary functional roles: the encoder-side TimesBlock introduces period-aware information after global contextualization, whereas the decoder-side TimesBlock provides a late residual refinement close to the denoising output. This design should not be interpreted as assuming that the before-last decoder position is universally optimal. A controlled ETTh-128 placement ablation in Sec. 5.4 compares the proposed configuration with E-only and an E + MidD variant having the same number of TimesBlocks. The results reveal a placement-dependent trade-off: the before-last decoder location favors correlational and predictive fidelity, whereas the middle-decoder location favors Context-FID and discriminative realism.

Selective insertion also limits repeated spectral processing. In our 3-encoder/3-decoder backbone, inserting TimesBlocks after all six Transformer blocks would triple the number of TimesBlock evaluations and substantially increase parameters and forward latency, as quantified in Sec. 5.4. We therefore retain E + before-last D as a selective late-refinement architecture rather than claiming universal optimality over all possible placements. The same architectural placement is used by PETS-U and PETS-F; the source and causal availability of period evidence, together with the generation-time control applied to the periodic residual, differ between the two protocols.

4.3. Unconditional Generation: PETS-U

Diffusion-PETS supports unconditional time-series synthesis through **PETS-U**, which models complete temporal trajectories without distinguishing observed and unknown regions. Accordingly, all temporal positions are available for period estimation, and no forecasting mask, history-derived period bank, or reconstruction guidance is introduced. PETS-U uses the shared learnable- k TimesBlock and PIB-Transformer described in Sec. 4.2, while adopting a batch-shared, full-sequence period-routing protocol appropriate for unconditional generation. Throughout this paper, **PETS-U** denotes the model using this batch-shared full-sequence routing and the canonical spectral-weighted aggregation. The final sampler, denoted **PETS-U-AA**, uses

the same frozen PETS-U checkpoint and applies Anti-Anchor symmetrically at the encoder- and decoder-side TimesBlocks during reverse diffusion.

4.3.1. Full-Sequence Batch-Shared Period Routing

Let $\mathbf{H} \in \mathbb{R}^{B \times T \times d}$ denote a latent representation entering a TimesBlock. Because unconditional generation has no causal boundary between historical and future regions, the complete temporal representation is used to estimate period evidence. The batch-level salience spectrum is

$$A^U(f) = \frac{1}{Bd} \sum_{b=1}^B \sum_{c=1}^d \left| \mathcal{F}_t(\mathbf{H}_{b,:c}) \right|, \quad f > 0, \quad (19)$$

from which the learnable- k dominant frequencies and corresponding periods are obtained as

$$\mathcal{I}_{\max}^U = \text{TopK}(A^U, k_{\max}), \quad \mathcal{P}_{\max}^U = \left\{ \left\lceil \frac{T}{f} \right\rceil \mid f \in \mathcal{I}_{\max}^U \right\}. \quad (20)$$

At TimesBlock $s \in \{E, D\}$, the checkpoint-learned count $k_s = \text{STECOUNT}(\alpha_s)$ activates the ranked prefix $\mathcal{P}_{\max}^{U,(s)} = \mathcal{P}_{\max}^U[1 : k_s]$ through the prefix-mask STE defined in Sec. 4.2.1. The candidate set is shared across the current batch, providing a consensus estimate of the temporal scales consistently represented across samples. Unlike the fixed history-derived bank used for forecasting, the unconditional period candidates are recomputed from the current latent representation at each TimesBlock evaluation and may therefore evolve along the reverse diffusion trajectory.

Batch-shared routing favors recurrent modes supported across the generated population and reduces sensitivity to transient sample-specific spectral peaks. This is well aligned with unconditional distribution learning, where the objective is to reproduce population-level temporal structure rather than condition on a particular observed trajectory. Its limitation is that sample-specific periodic modes with weak population support may receive less emphasis, which motivates controlling how the resulting period branches are combined during generation.

4.3.2. Unconditional Diffusion Training

PETS-U is trained on complete sequences using the same diffusion denoising objective as the shared Diffusion-PETS backbone. Given a clean sample $\mathbf{x}_0$, a noisy state is generated by the forward process,

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (21)$$

and the PIB-Transformer predicts the diffusion noise while its TimesBlocks extract period evidence from the complete latent sequence. The training objective is

$$\mathcal{L}_U = \mathcal{L}_{\text{diff}}(\mathbf{x}_{0,1:T}) + \lambda_{\text{FFT}} \mathcal{L}_{\text{FFT}}(\mathbf{x}_{0,1:T}), \quad (22)$$

where both terms are defined over the complete sequence. No temporal region is treated as an externally supplied

condition. During training, PETS-U retains the canonical spectral-weighted period-branch aggregation,

$$\mathbf{H}_{\text{per}} = \sum_{j=1}^{k_{\max}} w_j \mathbf{B}^{(j)}, \quad (23)$$

4.3.3. Unconditional Generation

The canonical TimesBlock aggregates all selected period branches using their normalized spectral weights. However, complementary branches may contain components that oppose the dominant periodic structure. To reduce such destructive interference during unconditional generation, the **PETS-U-AA** sampler applies *Anti-Anchor* aggregation at both the encoder- and decoder-side TimesBlocks. After training, the PETS-U checkpoint is frozen, and only the branch-aggregation operator is modified during unconditional reverse diffusion. This separation controls the generation-time interaction among already learned period branches without changing the period-discovery mechanism, retraining the denoising network, or introducing additional trainable parameters.

During generation, the reverse process starts from $\mathbf{x}_S \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. At each reverse step, the encoder- and decoder-side TimesBlocks recompute their batch-shared period candidates from their respective current full-length latent representations and construct the corresponding period-aligned 2D branches. The branches are then combined using Eq. (28) before the resulting periodic residual is fused back into the Transformer representation. Consequently, period discovery remains adaptive along the reverse trajectory, whereas the aggregation rule remains fixed at both period-aware locations.

Let $j^* = \arg \max_j w_j$ denote the branch with the largest period weight, and define $\rho_j = \max(p_j, p_{j^*}) / \min(p_j, p_{j^*})$. An active branch j is directly harmonically related to the dominant branch when

$$\left| \rho_j - \text{round}(\rho_j) \right| \leq \delta_h, \quad \delta_h = 0.05. \quad (24)$$

The direct harmonic family $\mathcal{H}$ contains j^* and all active branches satisfying Eq. (24). No transitive harmonic closure is used.

The weighted anchor-family and complementary contributions are

$$\mathbf{A} = \sum_{j \in \mathcal{H}} w_j \mathbf{B}^{(j)}, \quad \mathbf{C} = \sum_{j \notin \mathcal{H}} w_j \mathbf{B}^{(j)}. \quad (25)$$

With $W_A = \sum_{j \in \mathcal{H}} w_j$, $W_C = \sum_{j \notin \mathcal{H}} w_j$, and $W = W_A + W_C$, we define

$$\mathbf{A}^* = \frac{W}{W_A + \epsilon} \mathbf{A}, \quad \mathbf{R}_C = \mathbf{C} - \frac{W_C}{W_A + \epsilon} \mathbf{A}, \quad (26)$$

where $\epsilon > 0$ is a numerical stabilizer.

Anti-Anchor suppresses the component of the complementary residual that opposes the anchor. The projection is

Algorithm 1 PETS-U training and PETS-U-AA unconditional sampling

Require: PETS model f_θ , sequence length T , diffusion steps S

Training: PETS-U

- 1: **for** each complete sequence $\mathbf{x}_0$ **do**
- 2: Sample t, ϵ and obtain $\mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0)$
- 3: At each TimesBlock $s \in \{E, D\}$, compute $\mathcal{P}_{\max}^{U,(s)} \leftarrow \text{TopKPeriod}_{k_{\max}}(\mathbf{H}_{1:T,:}^{(s)})$
- 4: Set $k_s \leftarrow \text{STECCount}(\alpha_s)$ and $\mathcal{P}_{\max}^{U,(s)} \leftarrow \mathcal{P}_{\max}^{U,(s)}[1 : k_s]$
- 5: Construct period-aligned branches and apply the masked, renormalized aggregation in Eq. (13)
- 6: **end for**

Unconditional generation: PETS-U-AA

- 7: Freeze θ and initialize $\mathbf{x}_S \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 8: **for** $t = S, \dots, 1$ **do**
- 9: At each TimesBlock $s \in \{E, D\}$, recompute $\mathcal{P}_{\max}^{U,(s)}$ from its current full latent input
- 10: Construct the period-aligned branches and apply Eq. (28)
- 11: Apply the unconditional reverse diffusion update
- 12: **end for**
- 13: **return** generated sequence $\hat{\mathbf{x}}_0$

applied independently at each sample and temporal position along the feature dimension. For $\mathbf{a}, \mathbf{r} \in \mathbb{R}^d$, it is defined as

$$P_{\mathbf{a}}(\mathbf{r}) = \mathbf{r} - \frac{\min(0, \langle \mathbf{r}, \mathbf{a} \rangle)}{\|\mathbf{a}\|_2^2 + \epsilon} \mathbf{a}. \quad (27)$$

The resulting aggregation is

$$\mathbf{H}_{\text{AA}} = \mathbf{A}^* + P_{\mathbf{A}^*}(\mathbf{R}_C). \quad (28)$$

Thus, **PETS-U-AA** applies Eq. (28) at both period-aware locations during frozen unconditional sampling.

4.4. Conditional Forecasting: PETS-F

For probabilistic forecasting, we introduce a forecasting-specific variant of Diffusion-PETS, denoted as **PETS-F**. Unlike PETS-U, which is trained for unconditional generation using full-sequence period selection, PETS-F aligns the period-routing mechanism with the information available at forecast time. Let a complete training window be $\mathbf{X} = [\mathbf{X}^a, \mathbf{X}^b] \in \mathbb{R}^{L \times C}$, where $\mathbf{X}^a \in \mathbb{R}^{H \times C}$ denotes the historical part, $\mathbf{X}^b \in \mathbb{R}^{F \times C}$ denotes the forecasting part, and $L = H + F$. In our experiments, $L = 128$ and $H = F = 64$.

4.4.1. History-conditioned PETS-F Training

PETS-F retains the same PIB-Transformer architecture, diffusion objective, and spectral loss as the base Diffusion-PETS. The only task-specific modification is the source used for period-candidate routing. Given an intermediate TimesBlock input $\mathbf{Z} \in \mathbb{R}^{B \times L \times d}$, the period candidates for sample i are obtained only from its historical prefix,

$$\mathcal{P}_i^h = \text{TopKPeriod}(\mathbf{Z}_{i,1:H,:}), \quad (29)$$

while the selected period branches still transform the complete L -step representation $\mathbf{Z}_{i,1:L,:}$. The diffusion and spectral objectives are also evaluated over the complete training window,

$$\mathcal{L}_{\text{PETS-F}} = \mathcal{L}_{\text{diff}}(\mathbf{X}_{1:L}) + \lambda_{\text{FFT}} \mathcal{L}_{\text{FFT}}(\mathbf{X}_{1:L}). \quad (30)$$

Thus, history conditioning affects *which temporal scales are routed through the TimesBlock*, rather than restricting the generative training target to the historical region. This distinction is important: all complete windows used for training remain inside the chronological training partition, whereas the historical prefix provides causally matched history-only evidence that will be available at inference time.

This design reduces the train–inference mismatch that arises when period candidates are learned from a complete sequence during training but must be estimated from history alone during forecasting. Its main advantage is therefore causal alignment between training-time and inference-time period routing. The corresponding limitation is that PETS-F is a task-specific forecasting checkpoint rather than a single universal PETS-U checkpoint, and its routing policy is associated with the designated history length H .

4.4.2. Sample-Wise History Period Bank (SH64)

At a forecast origin o , only the observed history $\mathbf{x}_i^a = \mathbf{x}_{i,o-H:o}$ is available. SH64 computes a sample-specific spectrum from this history and fixes the resulting period bank throughout the reverse diffusion trajectory. Let $A_i(f)$ denote the non-DC Fourier amplitude averaged over variables and

$$P_i(f) = \frac{A_i(f)}{\sum_{f' > 0} A_i(f') + \epsilon}. \quad (31)$$

The $k_{\max}$ dominant frequency indices form the ranked bank $\mathcal{I}_{i,\max}^h = \text{TopK}_{k_{\max}, f > 0} P_i(f)$, from which the corresponding periods are obtained as

$$\mathcal{P}_{i,\max}^h = \left\{ \left\lceil \frac{H}{f} \right\rceil \mid f \in \mathcal{I}_{i,\max}^h \right\}. \quad (32)$$

Each TimesBlock $s \in \{E, D\}$ activates the prefix $\mathcal{P}_i^{h,(s)} = \mathcal{P}_{i,\max}^h[1 : k_s]$, where $k_s = \text{STECCount}(\alpha_s)$ is inherited from the trained checkpoint. The ranked bank and the two active prefixes remain fixed at all reverse steps. This sample-wise routing adapts PETS to the local temporal regime of each forecast instance and prevents future observations from influencing period selection. Fixing the bank also avoids noise-induced switching of candidate periods across reverse steps. However, because only a finite historical segment is available, its spectrum may be diffuse or unreliable when the local signal is weakly periodic or undergoing a regime transition.

4.4.3. Confidence-Gated Sampling (CG64)

To prevent uncertain period estimates from being enforced with the same strength as reliable ones, we augment SH64 with a parameter-free spectral confidence gate. For the normalized historical spectrum $P_i(f)$, we measure both the spectral mass captured by the selected candidates and the concentration of the complete spectrum:

$$M_{\text{bank}}^{(i)} = \sum_{f \in \mathcal{I}_{i,\max}^h} P_i(f),$$

$$H_{\text{spec}}^{(i)} = -\frac{\sum_f P_i(f) \log(P_i(f) + \epsilon)}{\log N_f},$$

$$c_i = M_{\text{bank}}^{(i)} \left(1 - H_{\text{spec}}^{(i)}\right). \quad (33)$$

A high c_i therefore indicates that a small number of dominant frequencies explain a large fraction of the observed spectral energy. To make the confidence scale comparable within each dataset, c_i is normalized using the 10th and 90th percentiles computed exclusively from the training partition:

$$\tilde{c}_i = \text{clip} \left(\frac{c_i - q_{10}^{\text{train}}}{q_{90}^{\text{train}} - q_{10}^{\text{train}} + \epsilon}, 0, 1 \right), \quad g_i = 0.25 + 0.75\tilde{c}_i. \quad (34)$$

The gate modulates the period-aware residual at the encoder-side and before-last-decoder TimesBlocks:

$$\Delta \mathbf{Z}_{\text{per}} = \text{TB}(\mathbf{Z}_{\text{base}}), \quad \mathbf{Z}_{\text{CG}} = \mathbf{Z}_{\text{base}} + g_i \Delta \mathbf{Z}_{\text{per}}. \quad (35)$$

When $g_i \approx 1$, the full period-aware transformation is retained; when the historical spectrum is uncertain, the model falls back toward the underlying Transformer representation while preserving a minimum periodic contribution. CG64 therefore addresses a limitation of SH64: the ranked-bank construction always returns period candidates even when no strongly supported periodic structure exists. Importantly, CG64 introduces no trainable parameters, does not modify the PETS-F checkpoint, and uses only the observed history and training-derived quantiles. Its main limitation is that a scalar confidence measure cannot perfectly distinguish weak but meaningful multi-period structure from spectrally ambiguous noise.

4.4.4. Conditional Reconstruction Guidance

SH64 and CG64 determine how period information is routed, whereas the observed values themselves are imposed through reconstruction-guided sampling. At reverse step t , let $\hat{\mathbf{x}}_{0,t}^a$ denote the historical part of the predicted clean sequence. We define

$$\mathcal{L}_{\text{rec},t} = D \left(\hat{\mathbf{x}}_{0,t}^a, \mathbf{x}^a \right), \quad (36)$$

and incorporate its gradient into the reverse diffusion update using the same reconstruction-guidance schedule as the conditional Diffusion-PETS sampler. The period bank and confidence gate are computed once from the observed history and remain fixed during this process. Consequently, neither period selection nor confidence estimation accesses the forecast target. Algorithm 2 summarizes the distinction between history-conditioned training and confidence-gated conditional sampling.

5. Experiment

5.1. Experimental Setup at a Glance

We evaluate Diffusion-PETS under two complementary settings: true-unconditional generation and conditional

Algorithm 2 History-conditioned PETS-F training and conditional sampling

Require: $L = H + F$, PETS model f_θ , candidate capacity k_{max}

Training: PETS-F

- 1: **for** each complete training window $\mathbf{X}$ **do**
- 2: Sample t and construct noisy state $\mathbf{X}_t$
- 3: **for** each TimesBlock $s \in \{E, D\}$ with input $\mathbf{Z}^{(s)}$ **do**
- 4: $\mathcal{P}_{i,\text{max}}^{h,(s)} \leftarrow \text{TopKPeriod}_{k_{\text{max}}}(\mathbf{Z}_{i,1:H}^{(s)})$ for each sample i
- 5: $k_s \leftarrow \text{STECCount}(\alpha_s)$; $\mathcal{P}_i^{h,(s)} \leftarrow \mathcal{P}_{i,\text{max}}^{h,(s)}[1 : k_s]$
- 6: Transform full $\mathbf{Z}_{i,1:L}^{(s)}$ using $\mathcal{P}_i^{h,(s)}$
- 7: **end for**
- 8: Update θ using the full-window diffusion and FFT losses
- 9: **end for**

Sampling: SH64/CG64 + reconstruction guidance

- 10: Observe $\mathbf{x}^a$ and compute $\mathcal{P}_{\text{max}}^h \leftarrow \text{TopKPeriod}_{k_{\text{max}}}(\mathbf{x}^a)$
- 11: **For** $s \in \{E, D\}$, set $k_s \leftarrow \text{STECCount}(\alpha_s)$ and $\mathcal{P}_{\text{max}}^{h,(s)} \leftarrow \mathcal{P}_{\text{max}}^h[1 : k_s]$
- 12: Set fixed g from Eqs. (33)–(34) for CG64; for SH64, $g = 1$
- 13: Initialize the unknown future region from diffusion noise
- 14: **for** $t = S, \dots, 1$ **do**
- 15: Evaluate PETS-F with fixed $\{\mathcal{P}^{h,(E)}, \mathcal{P}^{h,(D)}\}$ and g
- 16: Compute $\mathcal{L}_{\text{rec},t}$ on the observed history and apply the reconstruction-guided reverse update
- 17: **end for**
- 18: **return** generated future $\hat{\mathbf{x}}^b$

probabilistic forecasting. Both settings share the same period-aware Transformer backbone, while their period-routing and sampling policies are adapted to the information available in each task. Table 2 summarizes the shared architecture and task-specific experimental protocols.

The shared backbone consists of three encoder and three decoder layers with four attention heads, $d_{\text{model}} = 64$, and $d_{\text{ff}} = 128$. Period-aware processing follows the E + before-last D configuration, in which TimesBlocks are inserted after the encoder output and immediately before the final decoder block. Each TimesBlock identifies dominant temporal scales from the non-DC positive-frequency spectrum, constructs period-aligned 2D representations, and combines the resulting branches using the canonical spectral-weighted aggregation. All models use 500 diffusion steps and are trained for 18,000 optimization steps with a batch size of 128 and a learning rate of 10^{-4} .

For true-unconditional generation, PETS-U is trained on complete temporal windows using batch-shared full-sequence routing. The frozen checkpoint is evaluated as PETS-U-AA, with candidate periods recomputed from the current latent representation throughout reverse diffusion. We evaluate sequence lengths $T \in \{64, 128, 256\}$ using Context-FID, Correlational Score, Discriminative Score, and the TSTR-based Predictive Score.

For conditional forecasting, PETS-F is trained on complete 128-step windows drawn exclusively from the chronological training partition. At each TimesBlock, period candidates are selected sample-wise from the first 64 historical steps of the corresponding latent representation. The selected branches nevertheless process the complete 128-step representation, and both the diffusion and FFT losses

Table 2

Experimental details of Diffusion-PETS. PETS-U and PETS-F share the same period-aware backbone but differ in the source of period evidence and in their generation-time controls.

Item	Configuration
Shared backbone	
Architecture	Transformer encoder–decoder (3 encoder / 3 decoder layers, 4 heads, $d_{\text{model}} = 64$, $d_{\text{ff}} = 128$)
Period-aware module	E + before-last D placement; non-DC positive-frequency rFFT, learnable hard top- k selection, and period-aligned 2D reshaping
Optimization	500 diffusion steps; 18,000 optimization steps; batch size 128; learning rate 10^{-4}
Unconditional generation	
Data / length	ETTh ($C = 7$), Energy ($C = 28$), and AirQuality ($C = 9$); $T \in \{64, 128, 256\}$
Training aggregation:	Canonical spectral-weighted aggregation
Final unconditional sampling:	Frozen PETS-U-AA; period bank recomputed from the current full latent representation; no retraining or additional trainable parameters
Repetitions	3 model seeds (0, 42, 123) $\times$ 5 stochastic generation draws
Conditional forecasting	
Data / horizon	ETTh ($C = 7$), Energy ($C = 28$), and AirQuality ($C = 9$); 64 history + 64 future steps
Training routing	Complete 128-step windows from the chronological training partition; sample-wise period candidates extracted from each 64-step historical prefix
Training target	Selected branches process the complete 128-step representation; diffusion and FFT losses evaluated over the complete training window
Forecast routing	SH64 period bank computed once from each observed 64-step history and fixed throughout reverse diffusion; CG64 confidence gating; no future target used for routing
Conditioning / calibration	Causal reconstruction guidance; frozen mean-preserving spread calibration ($\gamma_{\text{cal}} = 1.5$) for the final predictive distribution
Repetitions	Development: 3 seeds $\times$ 5 draws; FINAL_TEST: 3 seeds $\times$ 10 draws; individual ablations report their own protocols

are evaluated over the full training window. Thus, history conditioning changes the source of period-routing evidence rather than restricting the generative training target.

At inference, the observed 64-step history defines a sample-specific SH64 period bank that remains fixed throughout reverse diffusion. CG64 scales the period-aware residual according to spectral confidence estimated from the observed history, while causal reconstruction guidance enforces consistency with the observed region. The final predictive distribution is adjusted using a frozen mean-preserving spread calibration with $\gamma_{\text{cal}} = 1.5$, which changes ensemble dispersion without changing its pointwise mean.

Forecast quality is assessed using ensemble-mean MAE and RMSE, expected-sample MAE, empirical CRPS, normalized Energy Score (nES), and Weighted Interval Score (WIS). Development experiments use three model seeds and five stochastic draws per seed, whereas the FINAL_TEST comparison uses three seeds and ten trajectories per seed. Ablation experiments that follow different repetition protocols state their settings explicitly in the corresponding table captions.

5.2. Unconditional Time-Series Generation

The established metrics in Table 3 preserve evaluator parity with prior work, while the structural and mechanism-specific analyses below provide complementary evidence. Table 4 reports the true-unconditional generation results

for PETS-U-AA, PETS-U, and Diffusion-TS across ETTh, AirQuality, and Energy. On ETTh, the relative benefit of period-aware modeling becomes more apparent as the sequence length increases. At $T = 64$, Diffusion-TS achieves the lowest Context-FID, Correlational, and Discriminative Scores and ties PETS-U-AA on the Predictive Score. At $T = 128$, PETS-U-AA achieves the lowest Context-FID, Correlational, and Discriminative Scores, while PETS-U and PETS-U-AA achieve the same lowest Predictive Score at the reported precision. At $T = 256$, PETS-U-AA retains the lowest Context-FID, whereas PETS-U achieves the lowest Correlational and Predictive Scores and Diffusion-TS remains marginally better on the Discriminative Score. Thus, PETS-U changes from being unfavorable at $T = 64$ to outperforming Diffusion-TS on three of the four metrics at both $T = 128$ and $T = 256$, supporting a sequence-length-dependent advantage on ETTh. PETS-U-AA further improves distributional alignment, particularly Context-FID, but does not uniformly improve all quality criteria.

AirQuality shows a different dependence on sequence length. At $T = 64$, PETS-U-AA achieves the lowest Context-FID and Correlational Scores, whereas Diffusion-TS achieves the lowest Discriminative Score and ties PETS-U for the lowest Predictive Score. At $T = 128$, the period-aware models provide their strongest overall result: PETS-U-AA achieves the lowest Context-FID and Correlational Scores, while PETS-U achieves the lowest Discriminative Score; PETS-U and PETS-U-AA tie for the lowest Predictive Score at the reported precision. In contrast, at $T = 256$, Diffusion-TS achieves the lowest value on all four metrics. This reversal indicates that increasing the sequence length does not by itself guarantee an advantage from explicit periodic modeling; rather, its effectiveness depends on whether the temporal structure exposed at a given window scale remains sufficiently stable and informative.

Energy constitutes a clearer adverse case for unconditional PETS-U. Diffusion-TS achieves lower Context-FID and Correlational Scores across all three sequence lengths and also retains the best Discriminative Score at $T = 128$ and $T = 256$. PETS-U nevertheless achieves the lowest Discriminative Score at $T = 64$ and the lowest Predictive Scores at $T = 128$ and $T = 256$, indicating that predictive temporal information can remain useful even when global distributional or dependency fidelity is weaker. PETS-U-AA consistently reduces Context-FID relative to PETS-U and often improves its correlational fidelity, but these gains are accompanied by trade-offs in discriminative or predictive performance. Overall, the results support a dataset- and scale-dependent interpretation of the proposed periodic inductive bias rather than universal superiority: the clearest gains occur on ETTh at medium and longer sequence lengths and on AirQuality at $T = 128$, whereas Energy and AirQuality at $T = 256$ expose regimes in which the baseline remains stronger.

Figs. 3 and 4 provide complementary qualitative and structural evidence for the unconditional generation results. As shown in Fig. 4, Diffusion-PETS exhibits stronger

Table 3

Metrics for unconditional time-series generation. ↓ indicates that lower values are better.

Metric	Formula	Description
Context-FID ↓	$S_{\text{CFID}} = \ \mu_r - \mu_g\ _2^2 + \text{tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$	Fréchet distance between real and generated TS2Vec embeddings; evaluates distributional similarity.
Correlational ↓	$S_{\text{corr}} = \frac{1}{10} \ \mathbf{R}_r - \mathbf{R}_g\ _1$	Difference between real and generated lag-zero channel-correlation matrices; evaluates dependency preservation.
Discriminative ↓	$S_{\text{disc}} = \text{Acc}(h_{r/g}) - 0.5 $	Distance of a real-versus-generated GRU classifier from chance accuracy; evaluates sample realism.
Predictive ↓	$S_{\text{pred}} = \text{MAE}(\mathbf{y}_r, f_g(\mathbf{X}_r))$	One-step MAE of a predictor trained on generated data and evaluated on real data under TSTR.

Notation: (μ_r, Σ_r) and (μ_g, Σ_g) are the embedding statistics of real and generated data; $\mathbf{R}_r$ and $\mathbf{R}_g$ are their lag-zero channel-correlation matrices; $h_{r/g}$ is the real-versus-generated classifier; and f_g is the predictor trained on generated data.

Table 4

Unconditional generation results on ETTh, AirQuality, and Energy at sequence lengths $T \in \{64, 128, 256\}$. PETS-U-AA denotes the frozen Anti-Anchor sampler, whereas PETS-U uses the canonical spectral-weighted aggregation. Results are reported as mean $\pm$ sample standard deviation over three model seeds (0, 42, and 123), with five stochastic generation draws per seed. All metrics are lower-is-better. Boldface denotes the best result among the compared generative models for each dataset-length pair. The rightmost column reports the one-step predictive score of a predictor trained and evaluated on real data under the same protocol and is shown only as a reference.

Dataset	T	Method	Context-FID ↓	Correlational ↓	Discriminative ↓	Predictive Score ↓	
						TSTR	Real-data ref.
ETTh	64	PETS-U-AA	0.455 $\pm$ 0.117	0.087 $\pm$ 0.015	0.099 $\pm$ 0.033	0.119 $\pm$ 0.000	0.117 $\pm$ 0.002
		PETS-U	0.732 $\pm$ 0.153	0.095 $\pm$ 0.019	0.134 $\pm$ 0.017	0.122 $\pm$ 0.001	
		Diffusion-TS	0.309 $\pm$ 0.018	0.075 $\pm$ 0.007	0.097 $\pm$ 0.002	0.119 $\pm$ 0.000	
	128	PETS-U-AA	0.856 $\pm$ 0.015	0.077 $\pm$ 0.003	0.145 $\pm$ 0.011	0.116 $\pm$ 0.003	0.118 $\pm$ 0.011
		PETS-U	0.929 $\pm$ 0.038	0.080 $\pm$ 0.007	0.154 $\pm$ 0.013	0.116 $\pm$ 0.001	
		Diffusion-TS	0.997 $\pm$ 0.769	0.094 $\pm$ 0.014	0.149 $\pm$ 0.045	0.118 $\pm$ 0.002	
	256	PETS-U-AA	1.606 $\pm$ 0.119	0.116 $\pm$ 0.013	0.216 $\pm$ 0.002	0.118 $\pm$ 0.006	0.112 $\pm$ 0.014
		PETS-U	1.663 $\pm$ 0.162	0.113 $\pm$ 0.006	0.207 $\pm$ 0.005	0.116 $\pm$ 0.002	
		Diffusion-TS	2.059 $\pm$ 1.070	0.148 $\pm$ 0.023	0.205 $\pm$ 0.051	0.118 $\pm$ 0.001	
AirQuality	64	PETS-U-AA	0.417 $\pm$ 0.027	0.083 $\pm$ 0.006	0.174 $\pm$ 0.030	0.023 $\pm$ 0.001	0.018 $\pm$ 0.000
		PETS-U	0.561 $\pm$ 0.200	0.111 $\pm$ 0.013	0.183 $\pm$ 0.028	0.022 $\pm$ 0.001	
		Diffusion-TS	0.477 $\pm$ 0.212	0.138 $\pm$ 0.032	0.164 $\pm$ 0.050	0.022 $\pm$ 0.001	
	128	PETS-U-AA	0.439 $\pm$ 0.091	0.090 $\pm$ 0.009	0.194 $\pm$ 0.040	0.020 $\pm$ 0.000	0.018 $\pm$ 0.001
		PETS-U	0.567 $\pm$ 0.104	0.110 $\pm$ 0.005	0.192 $\pm$ 0.030	0.020 $\pm$ 0.000	
		Diffusion-TS	0.640 $\pm$ 0.302	0.131 $\pm$ 0.025	0.195 $\pm$ 0.071	0.021 $\pm$ 0.000	
	256	PETS-U-AA	1.669 $\pm$ 1.194	0.141 $\pm$ 0.038	0.273 $\pm$ 0.027	0.021 $\pm$ 0.001	0.018 $\pm$ 0.001
		PETS-U	1.716 $\pm$ 1.264	0.142 $\pm$ 0.034	0.276 $\pm$ 0.015	0.021 $\pm$ 0.002	
		Diffusion-TS	1.047 $\pm$ 0.409	0.130 $\pm$ 0.029	0.246 $\pm$ 0.119	0.020 $\pm$ 0.000	
Energy	64	PETS-U-AA	0.904 $\pm$ 0.024	1.808 $\pm$ 0.195	0.322 $\pm$ 0.088	0.271 $\pm$ 0.021	0.246 $\pm$ 0.001
		PETS-U	0.938 $\pm$ 0.026	1.821 $\pm$ 0.191	0.299 $\pm$ 0.052	0.268 $\pm$ 0.013	
		Diffusion-TS	0.740 $\pm$ 0.099	1.321 $\pm$ 0.080	0.359 $\pm$ 0.021	0.262 $\pm$ 0.009	
	128	PETS-U-AA	1.494 $\pm$ 0.067	1.442 $\pm$ 0.073	0.445 $\pm$ 0.023	0.307 $\pm$ 0.021	0.245 $\pm$ 0.001
		PETS-U	1.629 $\pm$ 0.076	1.499 $\pm$ 0.086	0.428 $\pm$ 0.049	0.298 $\pm$ 0.026	
		Diffusion-TS	0.939 $\pm$ 0.054	1.339 $\pm$ 0.046	0.415 $\pm$ 0.024	0.312 $\pm$ 0.100	
	256	PETS-U-AA	1.949 $\pm$ 0.168	1.801 $\pm$ 0.285	0.465 $\pm$ 0.019	0.368 $\pm$ 0.058	0.243 $\pm$ 0.001
		PETS-U	2.077 $\pm$ 0.198	1.799 $\pm$ 0.034	0.478 $\pm$ 0.004	0.363 $\pm$ 0.038	
		Diffusion-TS	1.567 $\pm$ 0.361	1.210 $\pm$ 0.069	0.451 $\pm$ 0.035	0.386 $\pm$ 0.018	

Notes: TSTR denotes the one-step predictor trained on generated data and evaluated on real data. The real-data reference is trained and evaluated on real data under the same one-step protocol; The real-data reference is trained and evaluated on real data under the same one-step protocol; it is not a generative model and is excluded from the boldface comparison.

overlap with the real-data embeddings on both ETTh and AirQuality, while its marginal density closely follows the peak location, spread, and tail behavior of the original data; in comparison, Diffusion-TS shows more noticeable deviations in both the embedded geometry and density profiles. The structural-fidelity analysis in Fig. 3 further shows that both PETS-U variants reduce period-histogram divergence and turning-point error relative to Diffusion-TS. In particular, PETS-U-AA improves slope and curvature fidelity over PETS-U, with only small trade-offs in spectral and amplitude errors. Together, these results indicate that the period-aware representation improves not only global

distributional alignment but also the preservation of local temporal structure in unconditional generation.

5.3. Conditional Time-Series Generation

For each forecast origin, Diffusion-TS and PETS-F generate an ensemble of stochastic future trajectories. Accordingly, we evaluate not only the accuracy of the predictive center but also the fidelity of individual trajectories and the quality of the full predictive distribution. The primary forecasting metrics and their definitions are summarized in Table 5. Ensemble-mean MAE and RMSE are computed after averaging the stochastic trajectories at each forecast coordinate, thereby evaluating the pointwise accuracy of the

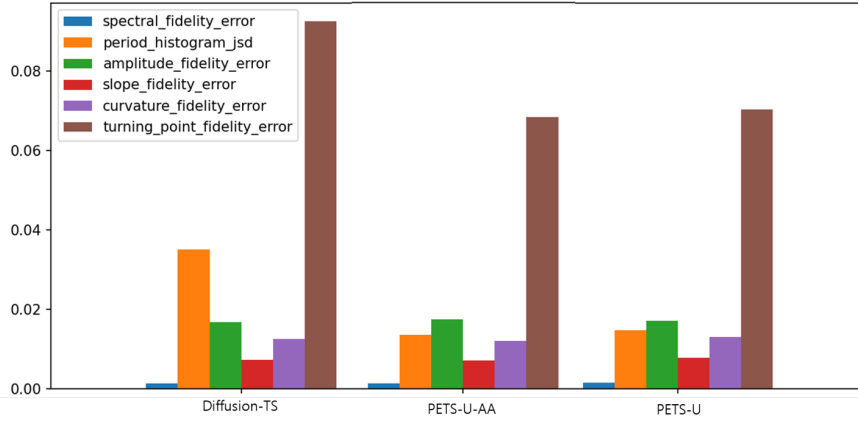


Figure 3: Secondary structural-fidelity errors on ETTh 128 for Diffusion-TS, PETS-U-AA, and PETS-U, shown from left to right. Both PETS-U variants reduce period-histogram divergence and turning-point error. PETS-U-AA further improves slope and curvature fidelity over canonical PETS-U, while slightly increasing spectral and amplitude errors. Lower values are better.

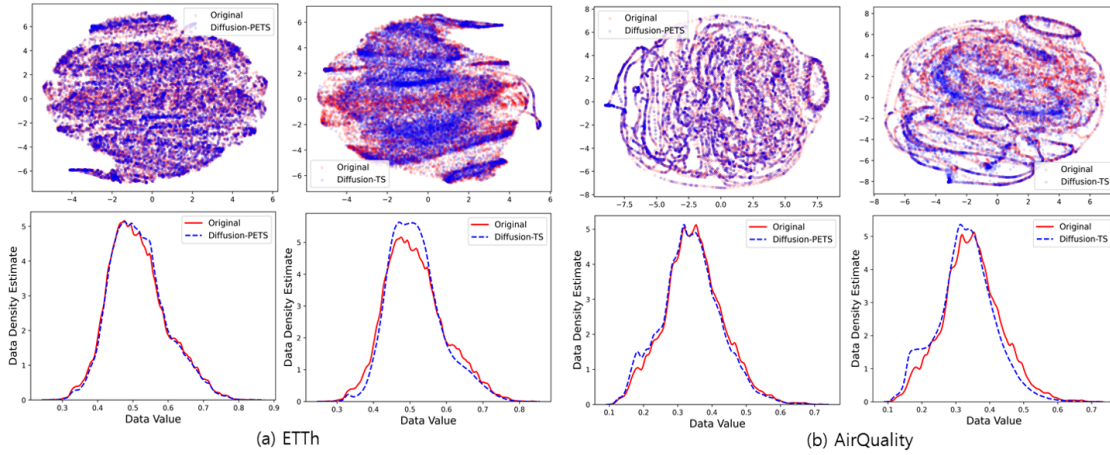


Figure 4: t-SNE visualization of real data and samples generated by PETS-U and Diffusion-TS. Top row: two-dimensional t-SNE projections of TS2Vec embeddings computed from complete real and unconditionally generated T -step sequences (red: real; blue: generated). Bottom row: kernel-density estimates of all normalized scalar values pooled across temporal positions and variables from the same real and generated sample sets.

predictive mean rather than an arbitrarily selected generated trajectory. This directly addresses the concern that sample-wise MAE or MSE alone may inadequately characterize probabilistic generation. Expected-sample MAE complements this measure by evaluating the expected error of an individual stochastic trajectory, while empirical CRPS, normalized Energy Score (nES), and Weighted Interval Score (WIS) assess marginal distributional quality, joint multivariate fidelity, and the calibration–sharpness trade-off, respectively. Together, these metrics prevent an accurate but overly smooth ensemble mean from concealing inaccurate or poorly calibrated individual forecast trajectories.

The contribution of the confidence-gated sampling policy is examined separately in Table 11. Starting from the same history-conditioned PETS-F checkpoints and SH64 period banks, CG64 modifies only the strength of the periodic residual according to confidence estimated from the

observed history. On the post-freeze FINAL_TEST evaluation, CG64 improves five of the six reported metrics on AirQuality, including reductions of 2.08% in ensemble-mean MAE, 1.95% in RMSE, 5.09% in expected-sample MAE, 0.44% in CRPS, and 2.26% in nES, although WIS increases slightly. On Energy, CG64 improves the central forecast accuracy but does not preserve the same advantage in the proper probabilistic scores, while ETTh exhibits a smaller mixed effect. These results suggest that the confidence gate is effective at suppressing unreliable period-aware corrections to the forecast trajectory, but may also reduce predictive dispersion in some temporal regimes. The final PETS-F system therefore combines CG64 with the frozen mean-preserving spread calibration to recover appropriate predictive uncertainty without changing the ensemble mean.

Table 6 compares this final PETS-F configuration with Diffusion-TS under the common FINAL_TEST protocol. On

Table 5

Primary metrics for probabilistic forecasting. ↓ indicates lower is better.

Metric	Formula	Description
Ensemble-mean MAE	$\frac{1}{N} \sum_{i=1}^N \bar{y}_i - y_i $	Pointwise error of the ensemble-mean trajectory; evaluates the accuracy of the predictive center.
Ensemble-mean RMSE	$\sqrt{\frac{1}{N} \sum_{i=1}^N (\bar{y}_i - y_i)^2}$	Ensemble-mean accuracy with greater sensitivity to large forecast errors.
Expected-sample MAE	$\frac{1}{MN} \sum_{m=1}^M \sum_{i=1}^N \hat{y}_i^{(m)} - y_i $	Expected pointwise error of a randomly sampled trajectory rather than only the ensemble mean.
Empirical CRPS	$\frac{1}{N} \sum_i \left[\frac{1}{M} \sum_m \hat{y}_i^{(m)} - y_i - \frac{1}{2M^2} \sum_{m,m'} \hat{y}_i^{(m)} - \hat{y}_i^{(m')} \right]$	Marginal proper score balancing agreement with observations and ensemble diversity.
Normalized Energy Score	$\frac{1}{B} \sum_n \left[\frac{1}{M} \sum_m \ \tilde{\mathbf{y}}_n^{(m)} - \tilde{\mathbf{y}}_n\ _2 - \frac{1}{2M^2} \sum_{m,m'} \ \tilde{\mathbf{y}}_n^{(m)} - \tilde{\mathbf{y}}_n^{(m')} \ _2 \right]$	Joint proper score over the full future horizon and all variables after training-statistic standardization.
Weighted Interval Score	$\frac{1}{N} \sum_i \frac{\frac{1}{2} \hat{q}_{0.5,i} - y_i + \sum_{a \in Q} \frac{a}{2} \text{IS}_{a,i}}{\frac{1}{2} + \sum_{a \in Q} \frac{a}{2}}$	Combines median error and central 50%, 80%, 90%, and 95% interval scores to assess calibration and sharpness.

Notation: $\bar{y}_i = M^{-1} \sum_m \hat{y}_i^{(m)}$. M , N , and B denote the numbers of stochastic trajectories, scalar forecast entries, and forecast windows, respectively. $\tilde{\mathbf{y}}$ denotes a raw-scale future vector standardized using training-partition statistics. $Q = \{0.5, 0.2, 0.1, 0.05\}$, and $\text{IS}_a = (u - l) + \frac{2}{a}(l - y)\mathbf{I}[y < l] + \frac{2}{a}(y - u)\mathbf{I}[y > u]$.

Table 6

Final-test comparison between Diffusion-TS and the final PETS-F forecasting configuration. PETS-F uses history-conditioned training, CG64 sampling, reconstruction guidance, and mean-preserving spread calibration ($\gamma_{\text{cal}} = 1.5$). Results are mean $\pm$ standard deviation over three model seeds, with 10 stochastic forecast trajectories per seed. All metrics are lower-is-better.

(a) Forecast accuracy					(b) Probabilistic forecasting quality				
Dataset	Method	MAE	RMSE	E-MAE	Dataset	Method	CRPS	nES	WIS
ETTh	Diffusion-TS	2.123 $\pm$ 0.069	3.862 $\pm$ 0.124	2.390 $\pm$ 0.130	ETTh	Diffusion-TS	1.741 $\pm$ 0.036	13.129 $\pm$ 0.161	7.005 $\pm$ 0.090
	PETS-F	2.057 $\pm$ 0.066	3.776 $\pm$ 0.249	2.392 $\pm$ 0.041		PETS-F	1.688 $\pm$ 0.077	12.480 $\pm$ 0.514	6.691 $\pm$ 0.482
Energy	Diffusion-TS	5.796 $\pm$ 0.097	20.158 $\pm$ 0.605	6.159 $\pm$ 0.115	Energy	Diffusion-TS	4.999 $\pm$ 0.055	25.824 $\pm$ 0.530	21.630 $\pm$ 0.243
	PETS-F	6.150 $\pm$ 0.224	22.131 $\pm$ 0.575	6.716 $\pm$ 0.292		PETS-F	5.440 $\pm$ 0.222	26.031 $\pm$ 1.529	23.549 $\pm$ 1.093
AirQ. [†]	Diffusion-TS	133.992 $\pm$ 7.841	250.957 $\pm$ 11.615	146.730 $\pm$ 9.720	AirQ. [†]	Diffusion-TS	107.746 $\pm$ 5.463	17.899 $\pm$ 0.534	434.388 $\pm$ 17.372
	PETS-F	133.673 $\pm$ 4.040	243.390 $\pm$ 9.831	153.577 $\pm$ 5.436		PETS-F	107.255 $\pm$ 3.165	17.312 $\pm$ 0.134	419.518 $\pm$ 16.286

Notes: PETS-F denotes the final history-conditioned PETS-F + CG64 + REC configuration followed by mean-preserving spread calibration. E-MAE denotes expected-sample MAE; nES denotes normalized Energy Score; and WIS denotes Weighted Interval Score. Boldface identifies the lower value within each dataset. [†]AirQuality is the held-out confirmatory FINAL_TEST, which was opened only after the method and calibration were frozen.

ETTh, PETS-F reduces ensemble-mean MAE and RMSE by 3.13% and 2.21%, respectively. The probabilistic scores also improve, with reductions of 3.07% in CRPS, 4.94% in nES, and 4.48% in WIS, while expected-sample MAE remains essentially unchanged relative to Diffusion-TS. The simultaneous improvement in the ensemble-mean errors and proper probabilistic scores indicates that the gain is not obtained merely by concentrating the predictive distribution around a smoother point forecast.

More importantly, PETS-F also improves the held-out confirmatory AirQuality FINAL_TEST, which was opened only after the forecasting method and calibration policy had been frozen. As shown in Table 6, PETS-F reduces ensemble-mean RMSE by 3.02%, CRPS by 0.46%, nES by 3.28%, and WIS by 3.42% relative to Diffusion-TS, together with a 0.24% reduction in ensemble-mean MAE. Expected-sample MAE is higher after spread calibration because the calibration deliberately increases the dispersion of individual trajectories around the unchanged ensemble mean. This trade-off is consistent with its intended role: CG64 controls

the period-aware forecast trajectory, whereas the subsequent calibration adjusts predictive dispersion.

The advantage is nevertheless dataset-dependent. On Energy, Diffusion-TS remains stronger than the final PETS-F configuration in both point-error and probabilistic metrics. We therefore do not interpret Table 6 as evidence of universal superiority across all temporal regimes. Rather, the results demonstrate that history-conditioned periodic modeling provides a confirmed forecasting advantage on the held-out confirmatory AirQuality test set, with complementary support from ETTh, while its benefit decreases when the periodic structure inferred from the observed history is less transferable to the future horizon.

5.4. Ablation Study

Anti-Anchor strategy comparison. Table 7 isolates the effect of Anti-Anchor aggregation at the encoder- and decoder-side TimesBlocks while keeping the period-candidate scope and FFT source fixed. Applying Anti-Anchor only to the encoder substantially improves the correlational score from

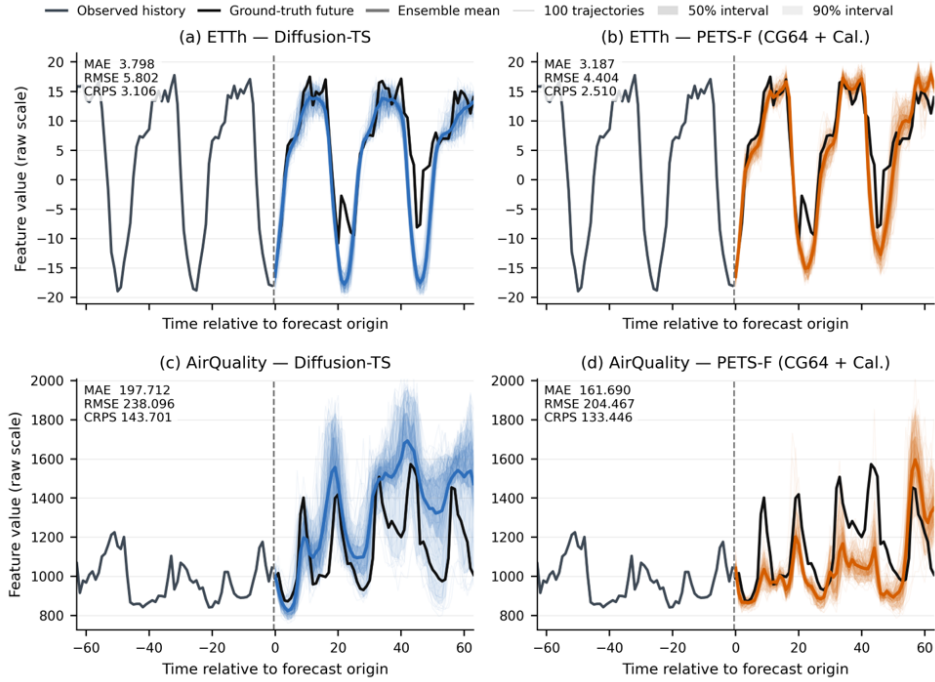


Figure 5: Qualitative probabilistic forecasts for identical observed histories, forecast origins, variables, and ground-truth futures. Each Diffusion-TS panel and each PETS-F panel overlays 100 stochastic trajectories, their pointwise ensemble mean, and the central 50% and 90% prediction intervals. For PETS-F, the trajectories are shown after applying the frozen mean-preserving spread calibration ($\gamma_{\text{cal}} = 1.5$); the ensemble mean is unchanged by this transformation. The displayed metrics are computed only for the illustrated windows, whereas the aggregate FINAL_TEST comparison over all windows and model seeds is reported in Table 6

0.0811 to 0.0753, while providing no improvement in Context-FID, suggesting that encoder-side suppression mainly affects multivariate dependency preservation. In contrast, applying Anti-Anchor only to the decoder markedly reduces Context-FID from 0.8844 to 0.8140 and slightly improves the discriminative score, although the correlational score increases. These complementary effects indicate distinct roles for the two TimesBlocks: the encoder-side intervention primarily regularizes cross-variable relations, whereas the decoder-side intervention has a stronger influence on global distributional realism.

Table 7 uses a five-seed protocol, whereas the representation and placement ablations below use their original three-seed canonical-aggregation protocol; absolute values across these tables should therefore not be compared directly. Applying the strategy at both TimesBlocks provides the most balanced improvement across the realism-oriented metrics. Compared with PETS-U, PETS-U-AA reduces Context-FID from 0.8844 to 0.8238, the correlational score from 0.0811 to 0.0772, and the discriminative score from 0.1404 to 0.1371. The Predictive score changes only marginally from 0.1161 to 0.1173. We therefore select PETS-U-AA as the balanced unconditional sampling configuration.

Representation strategy comparison. To examine whether the benefit of Diffusion-PETS arises specifically from period-aligned representation rather than from frequency information in general, we conducted a controlled comparison on ETTh with $T = 128$. As shown in Table 8, Period-Aligned 2D, STFT/Spectrogram, and Frequency-as-Channel use the same PETS-U backbone, E + before-last D placement, learnable hard top- k policy, training objective, period-routing scope, and evaluation protocol. The alternative modules were parameter-matched to the proposed model, with the total trainable parameter count differing by less than 0.2%. Thus, the primary experimental factor is the representation used to incorporate the selected spectral information.

Period-Aligned 2D achieves the best Correlational, Discriminative, and Predictive Scores, as well as the lowest Period-JSD. Relative to STFT/Spectrogram, it reduces Context-FID by approximately 15%, the Correlational Score by 26%, the Discriminative Score by 14%, and Period-JSD by 38%. Compared with Frequency-as-Channel, it reduces the Correlational and Discriminative Scores by approximately 38% and 10%, respectively, and Period-JSD by 65%. These results indicate that explicitly organizing latent values according to repeated-cycle and within-cycle coordinates is particularly effective for preserving the distribution of dominant temporal periods and their associated temporal dependencies.

Table 7

Encoder–decoder Anti-Anchor ablation under the true-unconditional generation protocol on ETTh with $T = 128$. All PETS-U variants use batch-shared period-candidate selection and full-sequence FFT, and differ only in the frozen branch aggregation applied at the encoder- and decoder-side TimesBlocks. Canonical denotes the spectral-weighted aggregation in Eq. (13), and AA denotes Anti-Anchor. For each model seed, the five generation draws are first averaged; the table reports mean $\pm$ sample standard deviation across the five independent model-seed means (0, 1, 2, 42, and 123).

Variant	Candidate scope	FFT source	Branch aggregation	Context-FID $\downarrow$	Corr. $\downarrow$	Disc. $\downarrow$	Pred. $\downarrow$
Diffusion-TS	Native	Native	Native	1.0611 ± 0.7048	0.1007 ± 0.0166	0.1531 ± 0.0484	0.1172 ± 0.0019
PETS-U	Batch-shared	Full sequence	Enc.: Canonical Dec.: Canonical	0.8844 ± 0.0648	0.0811 ± 0.0097	0.1404 ± 0.0163	0.1161 ± 0.0017
Encoder-AA	Batch-shared	Full sequence	Enc.: AA Dec.: Canonical	0.8948 ± 0.0526	0.0753 ± 0.0048	0.1411 ± 0.0169	0.1172 ± 0.0030
Decoder-AA	Batch-shared	Full sequence	Enc.: Canonical Dec.: AA	0.8140 ± 0.0636	0.0836 ± 0.0103	0.1396 ± 0.0127	0.1162 ± 0.0037
PETS-U-AA	Batch-shared	Full sequence	Enc.: AA Dec.: AA	0.8238 ± 0.0543	0.0772 ± 0.0054	0.1371 ± 0.0152	0.1173 ± 0.0032

Table 8

Controlled comparison of frequency-representation strategies on ETTh with sequence length $T = 128$ under the true-unconditional PETS-U protocol. All variants use the same Transformer backbone, E + before-last D placement, learnable hard top- k policy, training objective, and evaluation protocol. Results are mean $\pm$ sample standard deviation over three model seeds (0, 42, and 123), with five stochastic generation draws per seed. All metrics are lower-is-better; the best value in each column is bold. Spectral error is multiplied by 10^3 for readability. All representation variants use the canonical spectral-weighted aggregation so that representation is the only experimental factor.

Representation	Context-FID	Corr.	Disc.	Pred.	Period-JSD	Spectral error ($\times 10^3$)
Period-Aligned 2D	0.900 ± 0.046	0.078 ± 0.007	0.149 ± 0.014	0.115 ± 0.000	0.015 ± 0.002	1.463 ± 0.038
STFT / Spectrogram	1.059 ± 0.096	0.105 ± 0.012	0.174 ± 0.013	0.116 ± 0.001	0.024 ± 0.007	1.459 ± 0.096
Frequency-as-Channel	0.848 ± 0.131	0.126 ± 0.005	0.166 ± 0.017	0.118 ± 0.001	0.043 ± 0.004	1.408 ± 0.069

The comparison also reveals complementary strengths of the alternative representations. Frequency-as-Channel achieves the lowest Context-FID and spectral-fidelity error, while STFT/Spectrogram provides slightly better curvature and turning-point fidelity in the additional structural diagnostics. Therefore, the advantage of Period-Aligned 2D should not be interpreted as uniformly superior frequency reconstruction. Rather, its main benefit is the explicit cycle-phase organization of the latent sequence, which provides a favorable balance between period-distribution preservation, inter-variable dependency fidelity, sample realism, and downstream predictive utility in this controlled ETTh-128 setting. The comparison therefore supports a representation-specific quality trade-off rather than universal superiority of one frequency-domain encoding.

TimesBlock placement. We further examine the selective placement of TimesBlocks using a focused ETTh-128 PETS-U ablation. Three trained configurations are compared: encoder-only (E-only), encoder plus middle-decoder (E + MidD), and the proposed encoder plus before-last-decoder (E + LastBeforeD). E + MidD and E + LastBeforeD contain exactly the same two TimesBlocks and have identical parameter counts, thereby isolating the effect of the decoder-side insertion point.

As shown in Table 9, the placement comparison reveals a metric-dependent trade-off rather than a universally dominant location. Compared with E + MidD, E + LastBeforeD reduces the Correlational Score from 0.090 to 0.078 and the Predictive Score from 0.116 to 0.115, whereas E +

MidD achieves lower Context-FID (0.765 versus 0.900) and a lower Discriminative Score (0.128 versus 0.149). The same directional pattern is observed across all three model seeds. Thus, moving the decoder-side TimesBlock closer to the denoising output favors dependency preservation and downstream predictive structure, while earlier decoder insertion favors global distributional alignment and discriminative realism.

The E-only comparison further clarifies the contribution of the decoder-side periodic residual. Adding the before-last decoder TimesBlock improves the Correlational, Discriminative, and Predictive Scores relative to E-only, although Context-FID becomes higher. Accordingly, the decoder-side TimesBlock should be interpreted as a late period-aware refinement with a measurable quality trade-off, rather than as an operation that uniformly improves all generation criteria.

Finally, repeated all-layer insertion was evaluated only in terms of computational cost and was not trained or quality-ranked. Increasing the number of TimesBlocks from two to six raises the trainable parameter count from 1.592M to 3.888M and the measured denoiser-forward latency from 30.376 ms to 72.517 ms, corresponding to 2.44 $\times$ and 2.39 $\times$ increases, respectively. Peak memory increases more moderately from 232.461 MB to 248.183 MB. These measurements motivate selective insertion from a computational perspective, while no generation-quality claim is made for the untrained All-Layers configuration.

Period Candidate Selection for Forecasting. Table 10 isolates the effects of the FFT information source and

Table 9

Focused TimesBlock placement ablation on ETTh with $T = 128$ under the true-unconditional PETS-U protocol. Quality results are mean $\pm$ sample standard deviation over three model seeds (0, 42, and 123), with five paired generation draws per seed. E + MidD and E + LastBeforeD contain the same two TimesBlocks and differ only in the decoder-side insertion point. All-Layers was not trained and is included only for forward-cost profiling. All quality metrics are lower-is-better. All trained placement variants use the canonical spectral-weighted aggregation so that TimesBlock placement is the only quality factor.

(a) Generation quality						
Placement	#TB	Params (M)	Context-FID	Corr.	Disc.	Pred.
E-only	1	1.018	0.755 $\pm$ 0.025	0.096 $\pm$ 0.009	0.152 $\pm$ 0.009	0.116 $\pm$ 0.001
E + MidD	2	1.592	0.765 $\pm$ 0.075	0.090 $\pm$ 0.009	0.128 $\pm$ 0.002	0.116 $\pm$ 0.001
E + LastBeforeD	2	1.592	0.900 $\pm$ 0.046	0.078 $\pm$ 0.007	0.149 $\pm$ 0.014	0.115 $\pm$ 0.000
(b) Forward-cost profile						
Placement	#TB	Params (M)	Forward latency (ms)	Peak memory (MB)		
E-only	1	1.018	19.424 $\pm$ 0.698	232.563		
E + MidD	2	1.592	30.248 $\pm$ 0.871	236.871		
E + LastBeforeD	2	1.592	30.376 $\pm$ 0.714	232.461		
All-Layers (cost only)	6	3.888	72.517 $\pm$ 0.671	248.183		

Notes: Forward cost was measured with batch size 128, sequence length 128, FP32 execution, 20 warm-up forwards, and 100 timed forwards. All-Layers was instantiated only for cost profiling and was neither trained nor evaluated on generation-quality metrics.

Table 10

Ablation of period-candidate routing for PETS-F on ETTh. All variants use reconstruction-guided sampling and the canonical spectral-weighted aggregation. Lower values are better.

Model	Candidate scope	FFT source	Ensemble-mean MAE $\downarrow$	Ensemble-mean RMSE $\downarrow$	Expected-sample MAE $\downarrow$	Empirical CRPS $\downarrow$
PETS-F-BF	Batch-shared	Full sequence	0.0867	0.1209	0.0935	0.0734
PETS-F-BH	Batch-shared	History only	0.0834	0.1168	0.0895	0.0706
PETS-F-SH64	Sample-wise	History only	0.0825	0.1164	0.0885	0.0698

Notes: BF denotes batch-shared period candidates extracted from the full sequence; BH denotes batch-shared candidates extracted from the observed history; and SH64 denotes sample-wise candidates extracted independently from each 64-step observed history. The history-only setting modifies only TimesBlock candidate identities and candidate scores; the selected branches still transform the complete 128-step latent representation. All variants use the same diffusion objective, FFT loss, reconstruction guidance, and the canonical spectral-weighted aggregation. These values are obtained under the original ETTh routing-ablation protocol and should not be numerically compared with the separate FINAL_TEST table.

candidate-selection scope. Replacing full-sequence FFT with history-only FFT improves all four metrics under batch-shared routing, indicating that period candidates derived from the observed region are better aligned with the information available during forecasting. Changing the routing scope from batch-shared to sample-wise provides a further improvement, with SH64 achieving the lowest MAE, RMSE, expected-sample MAE, and CRPS.

Relative to BF, SH64 reduces ensemble-mean MAE, ensemble-mean RMSE, expected-sample MAE, and empirical CRPS by approximately 4.84%, 3.72%, 5.35%, and 4.90%, respectively. The comparison between BH and SH64 further shows that the gain is not explained solely by removing future-dependent FFT evidence; adapting the period candidates independently to each observed history also contributes to forecasting accuracy. These results motivate the history-conditioned, sample-wise routing policy subsequently used by PETS-F.

Table 11

Post-freeze FINAL_TEST ablation of confidence-gated sampling on the history-conditioned PETS-F backbone. Values denote the relative change from SH64 to CG64, averaged over three model seeds with 10 stochastic trajectories per seed. Negative values indicate improvement. This ablation was not used for method selection.

Dataset	Δ MAE	Δ RMSE	Δ E-MAE	Δ CRPS	Δ nES	Δ WIS
ETTh	-0.44%	+1.55%	-1.29%	+1.05%	-1.61%	+2.69%
Energy	-0.71%	-5.62%	-2.27%	+4.80%	+2.96%	+9.04%
AirQuality	-2.08%	-1.95%	-5.09%	-0.44%	-2.26%	+2.86%

Notes: SH64 and CG64 use identical history-conditioned PETS-F checkpoints, reconstruction guidance, forecast windows, and paired initial noise. Only the inference-time confidence gate differs. The relative change is computed as $\Delta = 100(\text{CG64} - \text{SH64})/\text{SH64}$. E-MAE denotes expected-sample MAE; nES denotes normalized Energy Score; and WIS denotes Weighted Interval Score. Boldface marks improvements. AirQuality was evaluated only after the final method had been frozen.

Table 12

Feature-level interaction between the Transformer representation and the TimesBlock periodic residual at the encoder- and decoder-side fusion sites. The magnitude ratio r measures the relative strength of the periodic residual, and cosine similarity c measures its directional alignment with the contextual feature.

Seed	Encoder-side fusion		Decoder-side fusion	
	Magnitude ratio r	Cosine similarity c	Magnitude ratio r	Cosine similarity c
0	0.6198	-0.2695	0.3701	-0.3132
42	0.6774	-0.2651	0.4581	-0.3881
123	0.6075	-0.2313	0.4313	-0.3224
Mean $\pm$ Std.	0.6349 ± 0.0373	-0.2553 ± 0.0209	0.4198 ± 0.0451	-0.3412 ± 0.0408

Notes: The statistics are computed on 256 fixed windows from the held-out ETTh test split using three independently trained PETS-F checkpoints. For each window, the magnitude ratio and cosine similarity are computed over the full 128×64 temporal feature representation at each fusion site and then averaged over the 256 windows for each model seed.

Table 11 isolates the effect of the confidence-gated CG64 policy after the history-conditioned PETS-F backbone and SH64 routing policy have been fixed. Because SH64 and CG64 use identical checkpoints, observed-history period banks, reconstruction guidance, forecast windows, and sampling conditions, the reported differences can be attributed specifically to the inference-time confidence gate. On AirQuality, CG64 improves ensemble-mean MAE, RMSE, expected-sample MAE, CRPS, and nES, with reductions of 2.08%, 1.95%, 5.09%, 0.44%, and 2.26%, respectively, while WIS increases by 2.86%. On Energy, the gate improves the three accuracy-oriented metrics, including a 5.62% reduction in RMSE, but degrades the proper probabilistic scores. ETTh shows a milder trade-off, with improvements in MAE, expected-sample MAE, and nES but small increases in RMSE, CRPS, and WIS. These results indicate that CG64 primarily suppresses period-aware corrections that are detrimental to the central forecast trajectory, whereas the resulting predictive spread may become too narrow in some regimes. This behavior motivates the subsequent mean-preserving spread calibration used in the final PETS-F predictive distribution. Importantly, this post-freeze ablation is used only to analyze the contribution of CG64 and was not used for method selection.

Computational complexity and efficiency. Diffusion-TS and Diffusion-PETS share the same Transformer backbone, whose self-attention cost scales as $\mathcal{O}((L_e + L_d)T^2d)$, where T is the sequence length and d the model width. Their main computational difference lies in the spectral modules. Diffusion-TS applies trend and top- k Fourier processing across decoder layers, resulting in an additional $\mathcal{O}(L_d T \log T)$ cost. In contrast, Diffusion-PETS invokes spectral processing only twice under the $E + \text{before-last } D$ configuration, with learnable top- k bounded by $2 \log T$. Thus, its additional spectral overhead remains independent of decoder depth, although each TimesBlock includes FFT-based period selection and two-dimensional feature extraction.

The empirical cost profile in Table 9 further quantifies this trade-off. On ETTh-128 with batch size 128, the proposed two-TimesBlock model uses 1.592M parameters

and 30.376 ms per denoiser forward, whereas the six-TimesBlock All-Layers configuration requires 3.888M parameters and 72.517 ms, corresponding to $2.44\times$ and $2.39\times$ increases, respectively. Because All-Layers was used only for cost profiling, no generation-quality comparison is made for this configuration.

5.5. Interpretability Analysis

Feature contribution analysis. We first quantify how the period-aware residual interacts with the Transformer representation at the two fusion sites. Let $\mathbf{H}^s$ denote the contextual feature before residual fusion and $\Delta\mathbf{H}_{\text{per}}^s$ the corresponding TimesBlock residual, where $s \in \{E, D\}$ denotes the encoder or decoder site. Their relative magnitude and directional alignment are measured as

$$r_s = \frac{\|\Delta\mathbf{H}_{\text{per}}^s\|_F}{\|\mathbf{H}^s\|_F + \epsilon}, \quad c_s = \frac{\langle \mathbf{H}^s, \Delta\mathbf{H}_{\text{per}}^s \rangle}{\|\mathbf{H}^s\|_F \|\Delta\mathbf{H}_{\text{per}}^s\|_F + \epsilon}. \quad (37)$$

As shown in Table 12, the encoder-side residual has a larger relative magnitude ($r = 0.6349 \pm 0.0373$) than the decoder-side residual ($r = 0.4198 \pm 0.0451$), whereas the decoder exhibits a more negative cosine similarity ($c = -0.3412 \pm 0.0408$ versus -0.2553 ± 0.0209). This indicates that the encoder introduces a stronger period-aware modification to the contextual representation, while the decoder applies a smaller but more directionally corrective late refinement. The consistently negative cosine similarities at both sites further show that TimesBlock does not merely reproduce the attention-derived features, but contributes a distinct periodic residual through additive fusion.

Denoising snapshots. To understand how Diffusion-PETS reconstructs time series, we analyze the conditional denoising process on representative ETTh (Fig. 6) and SensorDrift (Fig. 7) examples. Each figure visualizes (i) diagnostic period proposals obtained by applying the TimesBlock spectral analysis to the current decoder latent at selected reverse steps (top row), (ii) denoising snapshots of PETS-F and Diffusion-TS (middle rows), and (iii) the corresponding spectra (bottom row). These diagnostic proposals show how period evidence in the evolving decoder representation changes during denoising; they do not replace the fixed history-derived SH64 bank used by the final conditional sampler.

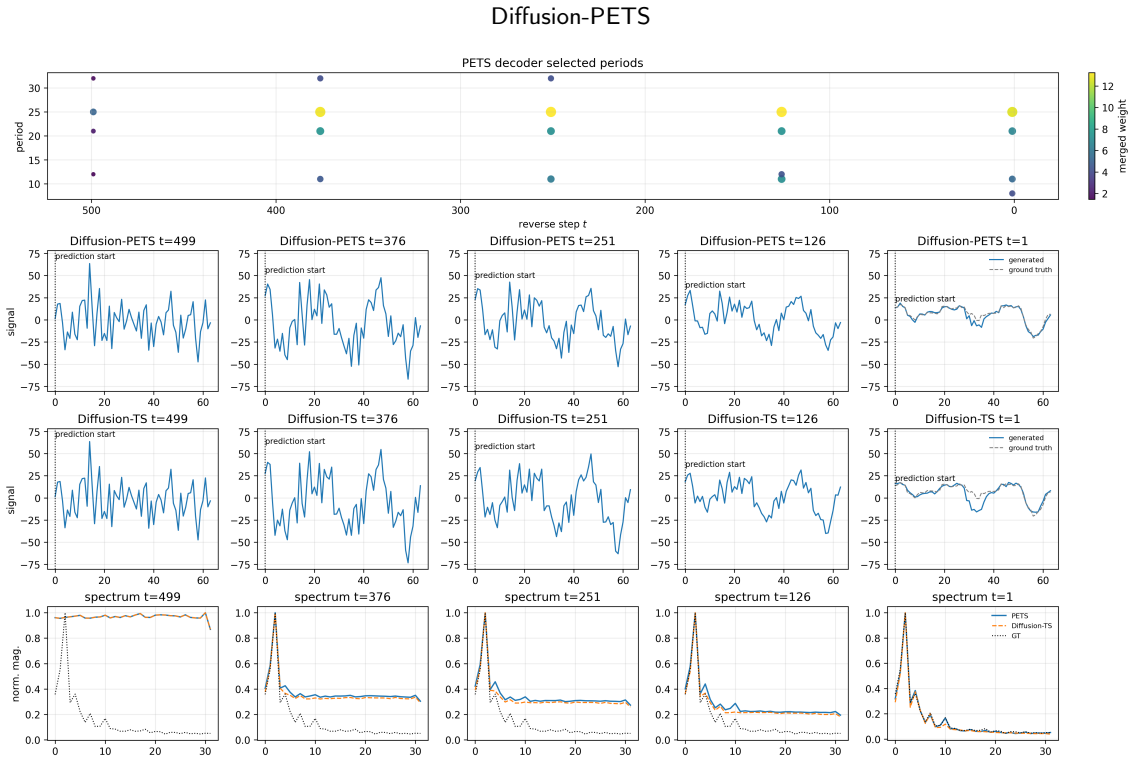


Figure 6: Interpretability of PETS-F on ETTh: diagnostic decoder-latent period proposals (top row), denoising snapshots of PETS-F and Diffusion-TS (middle rows), and the corresponding spectra (bottom row) across reverse steps.

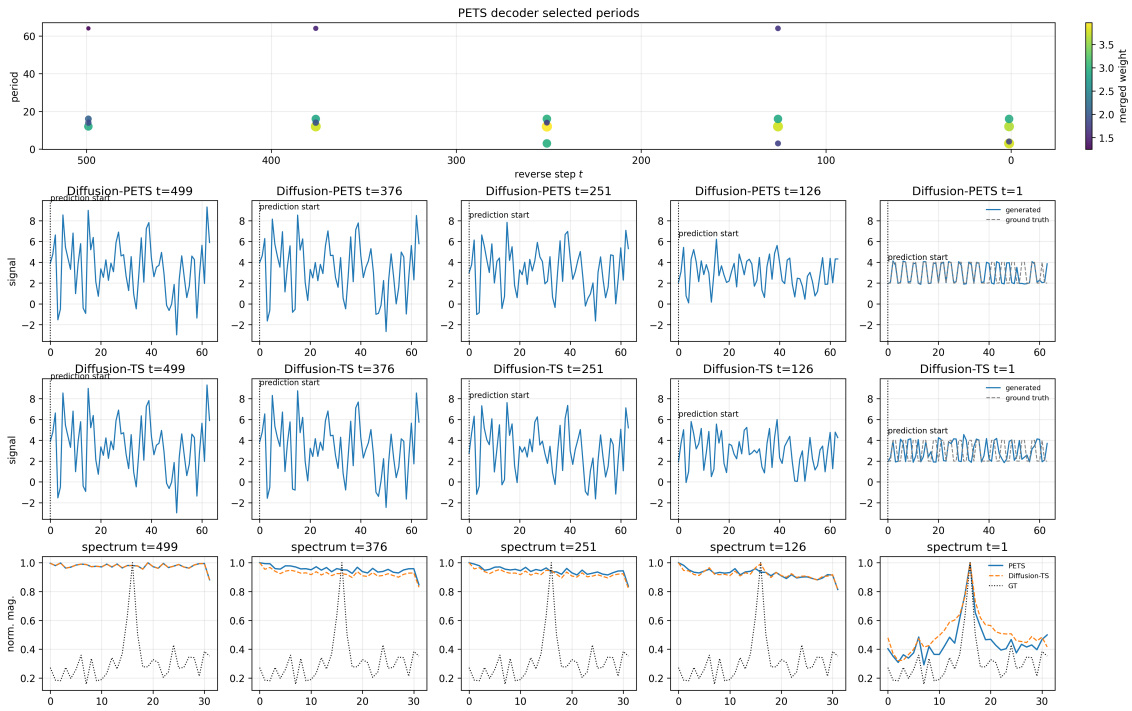


Figure 7: Interpretability of PETS-F on SensorDrift: diagnostic decoder-latent period proposals (top row), denoising snapshots of PETS-F and Diffusion-TS (middle rows), and the corresponding spectra (bottom row) across reverse steps.

On the ETTh example (Fig. 6), the top row shows that the spectral evidence extracted diagnostically from the decoder latent becomes increasingly concentrated around a small set of temporal scales as denoising proceeds, indicating that the model identifies consistent temporal scales. The middle rows

illustrate that the generated sequence evolves from noisy fluctuations ($t = 499$) to a clear oscillatory pattern at $t = 1$, closely matching the ground truth. In contrast, Diffusion-TS produces smoother but less structured outputs. The bottom row further confirms this behavior: the spectrum of

PETS increasingly aligns with the dominant frequency peak of the ground truth, whereas Diffusion-TS exhibits weaker alignment. Together, these observations show that PETS effectively recovers underlying periodic structure through both temporal and spectral consistency.

On the SensorDrift example (Fig. 7), a similar denoising progression is observed, but with a distinct period-selection pattern. As shown in the top row, the diagnostic decoder-latent period proposals remain more dispersed across reverse steps, indicating weaker and less concentrated temporal regularity compared to ETTh. Despite this, the middle rows show that Diffusion-PETS progressively reconstructs meaningful oscillatory structure, and at $t = 1$ its output aligns more closely with the underlying periodic behavior of the ground truth, whereas Diffusion-TS produces smoother signals with attenuated temporal variation. The spectral analysis in the bottom row further supports this observation: PETS captures the dominant frequency component more prominently than Diffusion-TS, although the peak remains broader and less sharply defined than in strongly periodic datasets. These results suggest that even under weaker or less stable periodicity, Diffusion-PETS can robustly recover dominant temporal patterns despite more distributed period selection.

These results demonstrate that PETS-F reconstructs time series through progressive refinement of latent temporal-scale evidence rather than purely pointwise denoising. The fixed SH64 bank provides the routing prior, while the evolving decoder representation and its diagnostic period proposals become increasingly aligned with the oscillatory ground-truth signals. Although the concentration of the diagnostic proposals varies across datasets, the overall behavior indicates that explicit period-aware modeling improves temporal structure preservation relative to Diffusion-TS.

6. Discussion

The results indicate that the benefit of explicit periodic modeling depends on both sequence length and the stability of temporal cycles. The unconditional results show clear gains in several settings but also reveal dataset- and metric-dependent trade-offs. In forecasting, PETS-F improves ensemble-mean and distributional metrics on ETTh and the held-out confirmatory AirQuality test set, whereas Diffusion-TS remains stronger on Energy. This contrast suggests that history-conditioned period routing is effective when the periodic evidence observed in the history remains informative over the future horizon, but is less reliable under weak periodicity or regime changes.

The task-specific protocols also show that period evidence should be adapted to the available information: PETS-U uses population-level full-sequence routing, whereas PETS-F uses causal, sample-wise history routing with confidence control. Nevertheless, PETS-F is tied to a designated history length and a forecasting-specific checkpoint, CG64 relies on a scalar confidence measure, and Anti-Anchor is currently a frozen inference-time intervention restricted to PETS-U;

it is not included in PETS-U training or PETS-F. Diffusion-PETS should therefore be regarded as a data-driven structural prior that complements, rather than replaces, model-based CPS approaches. Future work will investigate adaptive history lengths, branch-wise confidence control, learned aggregation, and faster diffusion sampling.

7. Conclusion

We introduced Diffusion-PETS, a diffusion-based Transformer that explicitly models interacting periodic structure through learnable hard top- k selection, period-aligned 2D branches, and selective encoder- and decoder-side residual fusion. The shared backbone supports true-unconditional generation through PETS-U and causal probabilistic forecasting through PETS-F using task-specific period routing and sampling.

Experiments demonstrate competitive unconditional generation performance with dataset- and length-dependent gains, together with forecasting gains on ETTh and the held-out confirmatory AirQuality test set; the Energy result confirms that the benefit is not uniform across temporal regimes. Feature-fusion and denoising analyses further indicate that TimesBlock contributes a distinct periodic residual rather than replacing the contextual representation learned by self-attention. These findings support explicit multi-period modeling as an effective inductive bias for uncertainty-aware sensor time-series generation.

Data and Code Availability

We use publicly available benchmark datasets widely adopted in time-series forecasting and generation. ETTh is obtained from the [Autoformer benchmark repository](#); The primary quantitative experiments use ETTh, Energy, and AirQuality. Weather is used only for the illustrative spectral comparison in Fig. 1, and SensorDrift is used only for the qualitative denoising analysis in Fig. 7. All datasets are publicly available for research and educational use; where explicit licenses are not specified, usage follows the terms and citation requirements of the original data providers. For datasets redistributed through third-party repositories, we follow the original retrieval procedures and acknowledge the source repositories to support reproducibility.

All datasets are used at their native temporal resolution without additional resampling. For true-unconditional generation, the MinMax scaler for each variable is fitted to the complete corpus used for model training. For conditional forecasting, all scalers are fitted exclusively to the chronological training partition and then applied unchanged to DEV_VAL and FINAL_TEST. In both settings, the normalized values are subsequently mapped to $[-1, 1]$ to match the zero-centered Gaussian diffusion process. The time series are segmented with unit-stride sliding windows; the unconditional sequence length is task dependent, whereas forecasting uses 128-step windows comprising 64 history and 64 future steps. The code is available at <https://github.com/parkjin-nim/Diffusion-PETS>.

Declaration of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors used an AI-assisted tool (ChatGPT) to improve language clarity and readability. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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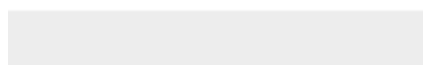
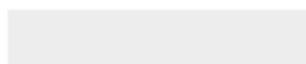






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